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Neural network model for liquefaction potential in soil deposits using Turkey and Taiwan earthquake data

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Abstract

Simplified methods have been practiced by researchers to assess nonlinear liquefaction potential of soil. Derived from several field and laboratory tests, various simplified procedures such as stress-based, strain-based, Chinese criteria, etc. have been developed by utilizing case studies and undisturbed soil specimens. In order to address the collective knowledge built up in conventional liquefaction engineering, an alternative general regression neural network model is proposed in this paper.

To meet this objective, a total of 620 sets of data including 12 soil and seismic parameters are introduced into the model. The data includes the results of field tests from the two major earthquakes that took place in Turkey and Taiwan in 1999 and some of the desired input parameters are obtained from correlations existing in the literature.

The proposed GRNN model was developed in four phases, mainly: identification phase, collection phase, implementation phase, and verification phase. An iterative procedure was followed to maximize the accuracy of the proposed model. The case records were divided randomly into testing, training, and validation datasets.

Generating a model that takes into account of 12 soil and seismic parameters is not feasible by using simplified techniques; however, the proposed GRNN model effectively explored the complex relationship between the introduced soil and seismic input parameters and validated the liquefaction decision obtained by simplified methods. The proposed GRNN model predicted well the occurrence/nonoccurrence of soil liquefaction in these sites. The model provides a viable tool to geotechnical engineers in assessing seismic condition in sites susceptible to liquefaction.
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Keywords: Liquefaction; General regression neural networks; GRNN; Kocaeli earthquake; Taiwan earthquake; Standard penetration test

1. Introduction

In soil deposits under undrained condition, earthquakes induce cyclic shear stresses, which may lead to soil liquefaction. This can be explained by the fact that in a liquefiable soil deposit, under dynamic load the pore water pressure increases, leading to a significant reduction in its shear strength and accordingly to liquefaction of the soil. In the literature, simplified methods are available for evaluating soil liquefaction potential using the in-situ tests results such as standard penetration testing (SPT) and cone penetration test (CPT) [1–5]

In the last two decades, artificial intelligence (AI) has been used in several applications in civil engineering, because of their heuristic problem-solving capabilities. Artificial neural network (ANN) is one of the AI approaches that can be classified as "machine learning". It has the ability to simulate the learning capabilities of the human brain by automating the process of knowledge acquisition and data mining. The ANN is a collection of interconnected computational elements called neurons that have performance characteristics similar to the biological neurons [6]. This brain-like structure makes ANN models superior to knowledge-based models and mathematical

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Notation

AI	artificial intelligence
ANN	artificial neural network
a_{max}	peak horizontal acceleration at ground surface
a_t	threshold acceleration
BPNN	back propagation neural network
CSR	cyclic stress ratio
$CRR_{7.5}$	cyclic resistance ratio for the magnitude 7.5 earthquakes
d_w	depth of ground water table
ER	hammer energy ratio
$F \leqslant 75\,\mu m$	percent fines less than 75 μm
GRNN	general regression neural network
MSF	magnitude scaling factor
M_v	earthquake magnitude
$(N_1)_{60}$	corrected standard penetration blow number
SNN	sequential neural network
V_s	shear wave velocity
z	depth of soil specimen
ϕ'	internal friction angle of soil
$\left(\frac{G}{G_{max}}\right)_t$	modulus reduction factor at the threshold strain, γ_t
σ	smoothing factor, or bandwidth
σ_{vo}	total vertical stress
σ'_{vo}	effective vertical stress
$\frac{\tau_{av}}{\sigma'_{vo}}$	cyclic stress ratio

formulae in making predictions that involve intuitive judgment and possess high degrees of nonlinearity.

Recently, ANN models were adopted by many researchers in the field of geotechnical engineering in solving varieties of problems. Goh [7] developed a back propagation neural network (BPNN) models to evaluate the shaft resistance of piles driven in cohesive soil. Teh et al. [8] developed a BPNN model to estimate the static pile capacity from the dynamic stress wave data. Abu Kiefa [9] developed a general regression neural network (GRNN) model to determine the capacity of piles driven in cohesionless soil. Kim and Kim [10] developed a BPNN model and a sequential neural network (SNN) model to predict the lateral load–deflection relationship of pile groups. Shahin et al. [11] developed an ANN model to predict the settlement of shallow foundations. Hanna et al. [12] developed an ANN to predict pile group efficiency.

Neural network methodology has been applied successfully to liquefaction potential; Goh [13,14] developed a BPNN model to assess soil liquefaction. He reported that neural networks were proven to be feasible tools for soil liquefaction assessments, simpler to apply and yield more reliable results when compared to conventional methodologies. It was noted that Goh has used relatively small dataset in developing the BPNN model, which considered the main limitation of ANN approaches. However, these models could be improved as new field case records became available.

Ural and Saka [15] developed a BPNN model based on liquefaction records of earthquakes from Japan, China, North and South America. In this research, case records were used in the developing stage. Juang et al. [16,17] developed ANN models to evaluate liquefaction resistance based on cone penetration test results and to predict liquefaction potential using standard penetration test results, respectively. Goh [18] developed probabilistic neural networks (PNN) models to analyze the databases based on the cone penetration test and shear wave velocity data. In this research, soil particle-size information was introduced into the architecture of the model as an input variable and the liquefaction phenomenon was considered as a classification problem.

These versatile applications make ANN models valuable problem-solving tools in the field of Geotechnical Engineering, which outperforms the conventional techniques in terms of accuracy and efficiency. In this study, the methodology of using general regression neural network, which is a well-recognized technique in solving and understanding many engineering problem, is applied to develop a predictive model for liquefaction triggering for soils under seismic loading. Unlike PNN and BPNN; GRNN, to authors' knowledge, has not been employed in soil liquefaction engineering for researching-layered soils. The authors believe that GRNN approach to soil liquefaction potential will contribute to the ongoing research in earthquake geotechnical engineering as it is capable of capturing complex interactions among the input variables introduced into the model.

In 1999, two major earthquakes, namely Chi-Chi, Taiwan earthquake (magnitude $M_w = 7.6$) and Kocaeli, Turkey earthquake (magnitude $M_w = 7.4$) triggered ground failure principally induced by soil liquefaction throughout the city of Adapazari (Turkey) and the cities of Wufeng, Nantou and Yuanlin (Taiwan). These case records were used extensively in this study to develop, train and validate the proposed GRNN model.

2. Overview of 1999 Kocaeli, Turkey earthquake and in-situ testing date

The epicenter of the Adapazari, Turkey earthquake was situated in the northwestern part of Turkey, in Izmit along the Northern Anatolian Fault (NAF), and affected a region with a population of approximately 15 million. The cause of the earthquake was a multiple rupture process in the 140 km long western part of the 1200 km long NAF. This natural disaster led to the loss of tens of thousands of people, and hundreds of thousands sustained injuries. In the vicinity of Adapazari peak ground accelerations were recorded at approximately 0.4 g. Displacements in the range of 3–4 m

were measured over a significant length of the fault. The maximum displacement was 5.1 ft immediately east of Arifiye. In the region, as many as 70% of the buildings were subjected to large ground settlements, liquefaction, or subsidence and sea water inundation [19]. Amplified ground motions on soft soil sites triggered liquefaction induced enormous structural damage in the region [20].

A collaborative research program was carried out by University of California—Berkeley, Brigham Young University, University of California—Los Angeles, ZETAS corporation, Sakarya University, Bogazici University and Middle East Technical University with the support of the US National Science Foundation, California Department of Transportation, California Energy Commission, and Pacific Gas and Electric Company in the region following the 1999 Kocaeli, Turkey earthquake. A total of 135 CPT profiles (of which 19 were seismic CPTs) and 46 soil borings with multiple SPT (often at 0.8 m spacing) were completed in the city of Adapazari. Details of this site investigation program are available at http://peer.berkeley.edu/turkey/adapazari/.

3. Overview of 1999 Chi-Chi, Taiwan earthquake and in-situ testing data

The epicenter located near to the town of Chi-Chi, had a very shallow depth of 7 km. Ground shaking exceeded 1.0 g in many places and triggered hundreds of strong motion instruments across the island. The Chelungpu Fault ruptured through hundred of structures and the earthquake generated thousands of landslides throughout the epicentral region. Damage was heaviest in the central counties of Taifung, Nantou and Yuanlin. The death toll surpassed 2400 and more than 10,700 people were injured. Over 8500 buildings were destroyed and another 6200 were seriously damaged, a majority of which were reinforced concrete structures with poorly designed columns that failed at the first floor. In the 1999 Chi-Chi, Taiwan earthquake, the seismic moment (a measure of the energy released by the earthquake) was 50% greater than the 1999 Kocaeli, Turkey Earthquake [21].

A series of site investigation programs were undertaken in 2000 by researchers with the National Center for Research in Earthquake (NCREE) in Taiwan and in 2001–2002 by the authors with funding from Pacific Earthquake Engineering Research Center [22]. The PEER and NCREE site investigation programs resulted in a total of 92 CPT soundings (of which 63 were seismic CPTs) and 98 soil borings with SPT (typically at 1.0 m spacing). The majority of the NCREE work was performed in the city of Yuanlin, whereas the entirety of the PEER work and some of the NCREE work was performed in the cities of Nantou and Wufeng. Results of both site investigation programs are synthesized at the address http://peer.berkeley.edu/lifelines/research_projects/3A02/ [23].

The dataset is interpreted from the 38 SPT borings carried out by the aforementioned sources for the Turkey Earthquake region and 25 SPT from the Taiwan earthquake region. Additionally, seismic cone penetration test results and spectral analysis of surface waves test (SASW) results are used to interpret shear wave velocity data.

4. Artificial neural network "ANN"

ANNs are advanced tools stimulated by the physical and computational characteristics of the human brain. Like biological neurons, they consist of interconnected information processing neural elements, neurons, working in union to make decisions, classifications, predictions, and forecasts. Neural networks are capable to learn linear and nonlinear functions that make them influential tools to analyze complex relations. Interconnections among neurons are established by weights, which are applied to all values passing through one neuron to another. Changing weights improve adaptabilities and prediction capabilities of the neural networks. Neural networks are arranged in three or more layers according to their connection to the outside world. Neurons receiving data are in the input layer; neurons carrying the results are in the output layer, and other neurons are in the hidden layers. Through the learning process, inputs and outputs data of a specific engineering problem are given and aforementioned weights among neurons are updated without requiring human development of algorithms. In the validation phase, the trained network is required to make predictions for a new set data that it has never been introduced during the previous phases. Neural network performs well as long as large volumes of data covering to all possible governing parameters and field conditions. Extensive information regarding the characteristics of neural networks methodology is outlined in the literature [24–26].

5. General regression neural networks "GRNN"

GRNN was introduced first by Specht [27] as an alternative to feed-forward neural networks. Unlike feed-forward neural networks, GRNN requires neither time-consuming trials nor alarming over-training conditions. Based on nonlinear regression theory, GRNN has special prediction capabilities for systems including sparse and noisy data. Unlike other kinds of neural networks, GRNN does not require initial setting of learning parameters, instead smoothing factor or bandwidth of all the parameters are required to be calculated. Therefore, GRNN training time is relatively short [28]. The success of the network depends on the smoothing factor that indicates how tightly the predictions match the actual values in the training patterns. Unlike PNN to categorizing data, GRNN is a universal tool to approximate to smooth functions and to produce continuous outputs with only one pass of training set. Being a supervised network, GRNN basically measures the distance of the training patterns in N-dimensional space, where N is the number of inputs, and estimates accordingly the output. GRNN has been widely covered [27,29,30].

6. "GRNN" model

The proposed GRNN model was developed in four phases; these are: identification phase, collection phase, implementation phase, and verification phase. An iterative procedure was followed to maximize the accuracy of the proposed model.

6.1. Identification phase

In this phase, soil and seismic parameters affecting soil liquefaction potential are identified. The primary concern in introducing 12 soil and seismic parameters into the model is to provide necessary information with reasonable accuracy for site characterization.

Soil behaves as a deformable body with increasing depth; hence depth is a site condition parameter and has a huge impact on soil response against its capacity to resist liquefaction. For liquefaction to occur, ground water conditions should be sufficient to create saturated soil conditions. Thus, depth to the groundwater table is an important consideration in identifying soils that are susceptible to liquefaction. Representing miscellaneous locations of soil deposits, depth of soil specimen and ground water table, which are real world parameters than can be obtained from various globally practiced field tests, are introduced into the proposed GRNN model.

SPT blow numbers have been commonly used for characterization of liquefaction resistance and SPT procedures have been developed by researchers to obtain information for liquefaction assessment. Thus, SPT penetration resistance is used as an index of liquefaction assessment model. Liquefaction potential of both cohesive and coarse-grained soils are influenced by fines content. As a compositional factor, fines content affects both cyclic shear strength and penetration resistance. Therefore, fines content is introduced into the model to improve the accuracy.

It has been proven that increase in overburden stresses increases susceptibility of soils to cyclic liquefaction. Representing geological setting of soil strata, overburden stress effects for liquefaction analysis procedures are investigated in the research of Boulanger [31] and Seed et al. [32]. Accordingly, total and effective overburden stresses are brought into the model. Shear wave velocity values are proven to represent the capacity of soil against liquefaction; therefore actual measured values of shear wave velocities are introduced into the network as an index of liquefaction resistance. Earthquake magnitude and maximum horizontal acceleration at ground surface characterize nature of loading, intensity of seismic ground shaking induced by the earthquakes. Obtained from strong ground networks, they are real-world parameters that are crucial to introduce into the seismic model.

Conventional liquefaction potential assessment procedures profoundly rely on empirical correlations such as cyclic stress ratio for stress-based methodology and threshold acceleration for strain-based methodology. Accordingly, parameters derived from field tests are used directly through empirical calculations. These correlations derived from case histories and laboratory experiments yield reliable results. Dobry et al. [33] showed that liquefaction resistance of saturated undrained specimens against soil liquefaction can also be quantified by threshold shear strain, which is not dependent on the method of sample preparation, and it is approximately 0.01%. By utilizing this value, acceleration corresponding to threshold shear strain value is evaluated by the strain-based assessment procedure for liquefaction occurrence. Strain-based liquefaction potential assessment procedure is idealized for sand grains and then generalized for natural soil deposits. Threshold acceleration value together with strain-based factor of safety states the exceedance of the threshold strain which is required for liquefaction to occur. In order to evaluate the threshold acceleration, easily obtained field tests variables and earthquake characteristics are utilized. Therefore, threshold acceleration can be considered as a real-world parameter which contributes strain-based procedure knowledge into the complex liquefaction problem. Moreover, formulated by Seed et al. [1], cyclic stress ratio is a normalized measure of cyclic load severity and it represents the seismic demand on soil to liquefy. In that respect, threshold acceleration and cyclic stress ratio values are introduced into the model.

The angle at which soil resists shearing is defined as internal friction angle. Thus, internal friction angle illustrates shearing strength. With the advent of field testing methods, friction angle can easily consistently be estimated from SPT, CPT tests. Internal friction angle of soil is introduced into the model by using correlations based on SPT penetration blow counts in order to assess most realistic values.

6.2. Collection phase

The field test results of the two earthquakes took place in Chi-Chi, Taiwan (1999) and Kocaeli, Turkey (1999) are used in this investigation to develop the proposed GRNN model. The dataset consists of 620 case records; 330 for the Kocaeli earthquake and 290 for the Taiwan earthquake. The dataset represents 364 cases that liquefied and 256 cases that did not liquefy.

The database covers a wide spectrum of soil and seismic parameters because in this paper, stress-based and strain-based liquefaction evaluation procedures are used to predict liquefaction potential. Furthermore, liquefaction probability of soils with significant fines is tested with so-called Chinese criteria. In the literature, among various simplified techniques, four techniques based on: (i) standard penetration test, (ii) cone penetration test, (iii) shear wave velocity measurements, and (iv) Becker penetration test are commonly practiced by researchers [5]. The authors believe that better predictions can be achieved by employing all the governing soil and seismic

parameters in one model. Therefore, in this study, instead of following only one type of these aforementioned criteria and incorporating a limited number of soil and seismic variable, a comprehensive approach that consists of all independent soil and seismic parameters proven to be influential for decision making are taken into account. In that respect, the authors believe that it is valid to include input variables coming from different concepts. On the other hand, presenting alternative parameters to the model improves prediction capabilities of neural networks that have thought-provoking parameter recognition capabilities.

The database, given in Table 2, consists of soil and seismic parameters including depth of the soil layer (z), corrected standard penetration blow numbers ($N_{1,60}$), percent finest content less than 75 μm ($F \leqslant 75$ μm, %), depth of ground water table (d_w), total and effective overburden stresses (σ_{vo}, σ'_{vo}), threshold acceleration (a_t), cyclic stress ratio (τ_{av}/σ'_{vo}), shear wave velocity (V_s), internal friction angle of soil (ϕ'), earthquake magnitude (M_v), and maximum horizontal acceleration at ground surface(a_{max}).

In Table 2, the depths of the soil specimen, given in the first column, were taken up to 20 m from the ground surface. The standard penetration test results, given in column 2, were corrected according to the procedure described by Youd et al. [5]. Hammer energy ratios (ER) for Kocaeli earthquake case records are the actual values measured in the field. However, the "ER" values for Taiwan earthquake case records were based on strain gauge measurements due to failing accelerometers. Therefore, the "ER" values reported by Stewart et al. [23] were adjusted to 69% for depths up to 3 m and 75% for depths below 3 m. Column 3 presents the percentage finest content less than 75 μm, which are the actual values measured in the field. Finest content is also taken into consideration while calculating the cyclic resistance ratio ($CRR_{7.5}$), due to its notable influence on soil liquefaction [2]. The depth of the ground water table is given in column 4. Total and effective overburden stresses are presented in column 5–6, respectively. The threshold acceleration values, listed in column 7, were calculated following the procedure given by Dobry et al. [33,34]. In column 8, cyclic stress ratios required to generate soil liquefaction were tabulated. In column 9, the shear wave velocities, which were the actual measured values from related seismic cone test (SCPT) and spectral analysis of surface waves test (SASW) were presented. Soil internal friction is presented in column 10. Earthquake magnitudes of all the soil data obtained from Turkey and Taiwan earthquake regions are given in column 11. Maximum horizontal accelerations at ground surface are given in the column 12. The distribution of the peak accelerations recorded by strong motion station is given in Table 1. In Table 1, the closest measurements of all available strong ground motion recorders are taken into account in order to characterize the region realistically.

Table 1
Maximum horizontal accelerations recorded by strong motion stations

Area	Taiwan			Turkey
Location	Nantou	Wufeng	Yuanlin	Adapazari city
Value	0.38 (g)	0.67 (g)	0.18 (g)	0.4 (g)
Station	TCU076	TCU065	TCU110	NSMP[a]

[a]National strong motion network.

7. Liquefaction evaluation

This study explores liquefaction potential of soil deposits. It is not possible to investigate liquefaction potential of soil strata at particular depth by field observations. Thus, conventional methods are used to calculate the possibility of liquefaction at each depth. The authors believe that SPT test results supported by shear wave velocity measurements are capable of accurately identifying and predicting liquefaction potential of liquefiable layers. Accordingly, liquefaction potential of soil layers has been evaluated by means of an assortment of empirical criteria which sometimes found to be more reliable than analytical calculations and frozen sampling. Given in column 13 in Table 2, liquefaction susceptibility of soils in Turkey and Taiwan earthquake regions are determined by the three following criteria: (i) stress-based liquefaction triggering analyses developed for SPT tests [5], (ii) strain-based procedure [33], and (iii) Chinese criteria [35].

Seed et al. [2] correlation, modified by Youd et al. [5], is controlled by mostly clean sands and silty sands with less than 35% fines; however, many of case histories from both earthquake regions involve high cyclic stress ratios and high finest content soils. Therefore, three different soil liquefaction potential evaluation procedures are used. From an earthquake geotechnical engineer's point of view, these earthquakes provide an exceptional opportunity to advance the development of empirical liquefaction assessment methodology in that case histories in the city of Adapazari (Turkey) and the cities of Wufeng, Nantou and Yuanlin (Taiwan) [23] and complete the necessary parameter spaces in the well known Seed et al. [2] procedure.

Based on the procedure given by Youd et al. [5], relationship between cyclic stress ratio and corrected standard penetration resistance for sands and silty sands with varying peak ground acceleration and earthquake magnitudes are given in Figs. 1–4.

Moreover, soils with significant plasticity are evaluated for liquefaction susceptibility by the Chinese Criteria proposed by Finn et al. [35]. This criterion specifies that soils with plastic fines can liquefy if all the four following conditions are met: (1) percent finer than 0.005 mm $\leqslant$20%, (2) (liquid limit + 1%) $\leqslant$35%, (3) (water content + 2%) $\geqslant$0.9* (liquid limit + 1%), and (4) liquidity index (based on liquid limit + 1% and water content + 2%) $\leqslant$0.75 [36]. The graphical representation of this criterion with soil data from both earthquakes are shown in Fig. 5–7.

Table 2
Summary of SPT liquefaction case records

No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Parameters	Z	$(N_1)_{60}$	$F \leqslant 75\,\mu m$	d_w	σ_{vo}	σ'_{vo}	a_t	τ_{av}/σ'_{vo}	V_s	ϕ'	M_v	a_{max}	Liquefaction	GRNN
Unit	(m)	—	(%)	(m)	(kPa)	(kPa)	(g)	—	(m/s)	(°)	—	(g)	—	—
Summary of training phase	7.0	19	70	0.77	127.9	65.6	0.13	0.45	250	33.42	7.4	0.40	Yes	Yes
	3.9	15	5	1.68	70.0	47.8	0.12	0.35	185	31.49	7.4	0.40	Yes	Yes
	8.4	31	7	0.37	148.0	67.7	0.04	0.49	150	38.00	7.4	0.40	No	No
	9.2	7	94	1.65	174.1	98.6	0.04	0.39	160	29.17	7.4	0.40	Yes	Yes
	13.2	18	21	0.60	250.5	126.9	0.06	0.19	227	33.86	7.6	0.18	No	No
	5.8	9	57	0.80	98.2	49.1	0.05	0.23	145	28.49	7.6	0.18	Yes	Yes
	13.2	42	42	1.90	269.7	158.9	0.15	0.35	363	42.18	7.6	0.38	No	No
	4.2	9	54	0.75	82.5	48.7	0.05	0.73	130	28.49	7.6	0.67	Yes	Yes
Summary of testing phase	4.1	7	97	0.82	74.1	41.3	0.02	0.43	85	28.49	7.4	0.40	Yes	Yes
	7.4	20	1	1.50	138.1	79.1	0.06	0.40	180	35.49	7.4	0.40	No	No
	8.0	7	99	0.54	148.4	73.8	0.03	0.45	134	28.48	7.4	0.40	Yes	No[a]
	3.2	5	70	1.78	52.9	39.0	0.10	0.16	158	26.00	7.6	0.18	Yes	Yes
	17.2	30	20	0.99	339.1	180.0	0.10	0.33	315	39.29	7.6	0.38	No	No
	11.8	26	51	1.40	236.6	134.6	0.04	0.67	182	36.97	7.6	0.67	No	No
	11.2	13	40	0.50	226.6	121.7	0.08	0.72	239	31.49	7.6	0.67	Yes	Yes
	5.6	26	36	1.57	112.9	73.4	0.05	0.65	145	35.10	7.6	0.67	No	No
Summary of validation phase	10.0	25	6	0.78	186.6	94.4	0.09	0.43	250	37.32	7.4	0.40	No	No
	11.0	33	10	0.82	211.1	109.3	0.35	0.41	500	46.35	7.4	0.40	No	No
	2.9	7	59	0.50	52.9	28.9	0.02	0.45	67	27.75	7.4	0.40	Yes	Yes
	4.2	5	99	1.78	72.0	48.2	0.08	0.17	158	26.00	7.6	0.18	Yes	Yes
	2.2	9	15	2.00	39.1	37.1	0.13	0.12	150	28.49	7.6	0.18	Yes	Yes
	11.8	23	18	2.00	228.5	132.3	0.05	0.18	200	35.87	7.6	0.18	No	No
	14.2	36	19	0.70	320.6	188.2	0.06	0.34	242	41.35	7.6	0.38	No	No
	2.8	9	31	1.57	55.2	43.1	0.07	0.55	122	28.49	7.6	0.67	Yes	Yes

[a]These case records indicate errors in the architecture.

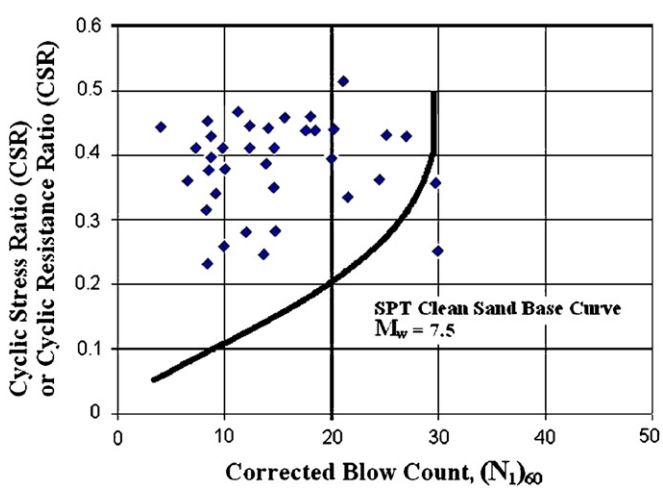

Fig. 1. Relationship between cyclic stress ration (CSR) and corrected standard penetration test (SPT) number for Kocaeli, Turkey Earthquake data for the peak ground acceleration (a_{max}) of 0.4 g and a moment magnitude (M_w) of 7.4 (Modified for the magnitude scaling factor given by Andrus and Stokoe [39]).

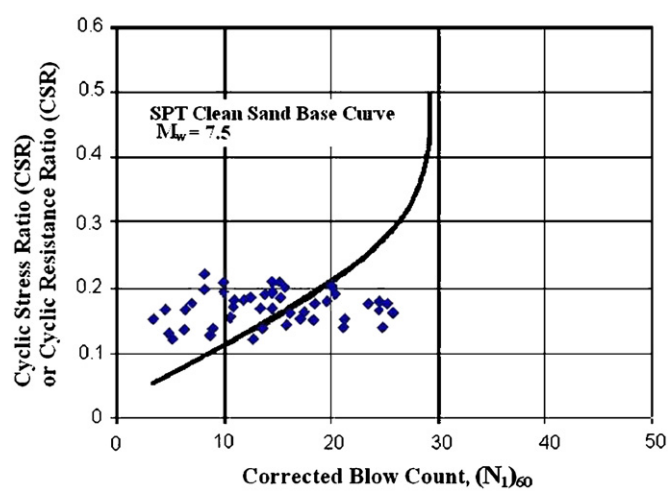

Fig. 2. Relationship between cyclic stress ration (CSR) and corrected standard penetration test (SPT) number for Chi-Chi, Taiwan Earthquake data for the peak ground acceleration (a_{max}) of 0.18 g and a moment magnitude (M_w) of 7.6 (Modified for the magnitude scaling factor given by Andrus and Stokoe [39]).

Given in Eq. (1), the threshold acceleration can be calculated as follows:

$$\frac{a_t}{g} = \frac{V_s^2 \left(\frac{G}{G_{max}}\right)_t \gamma_t}{0.65 g z r_d}, \tag{1}$$

where V_s is the shear wave velocity, γ_t is the threshold strain, $(G/G_{max})_t$ is the modulus reduction factor at the threshold, r_d is the stress reduction coefficient, and z is the depth of soil specimen. The modulus reduction factor at the threshold, $(G/G_{max})_t$, was assumed to be 0.8 with a strain level of an order of 0.01% [37]. Factor

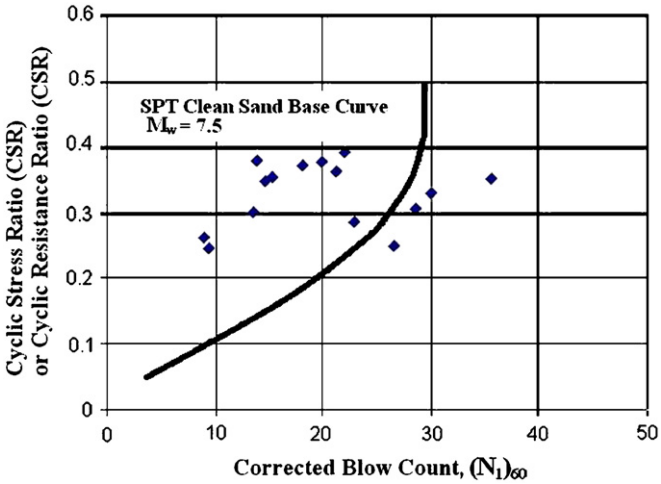

Fig. 3. Relationship between cyclic stress ration (CSR) and corrected standard penetration test (SPT) number for Chi-Chi, Taiwan Earthquake data for the peak ground acceleration (a_{max}) of 0.38 g and a moment magnitude (M_w) of 7.6 (Modified for the magnitude scaling factor given by Andrus and Stokoe [39]).

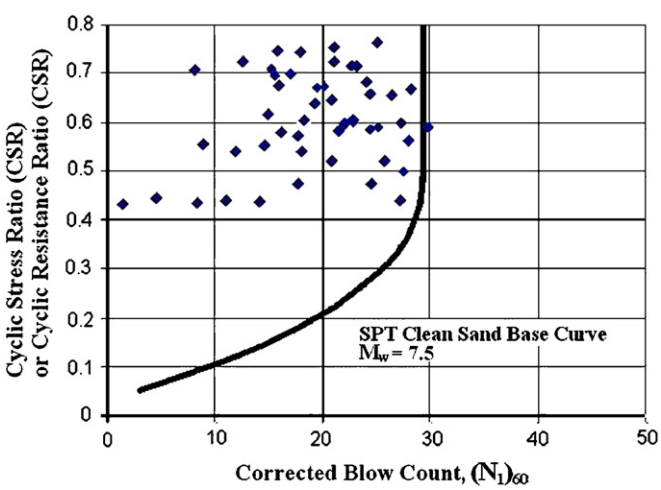

Fig. 4. Relationship between cyclic stress ration (CSR) and corrected standard penetration test (SPT) number for Chi-Chi, Taiwan Earthquake data for the peak ground acceleration (a_{max}) of 0.67 g and a moment magnitude (M_w) of 7.6 (Modified for the magnitude scaling factor given by Andrus and Stokoe [39]).

of safety in threshold acceleration criteria is given in Eq. (2) below

$$F_a = \frac{a_t}{a_{max}}, \tag{2}$$

$F_a > 1$ implies that there is no risk, $F_a \leqslant 1$ not necessarily mean that liquefaction will occur; instead, the value predicts that there will be gross sliding of the grain-to-grain contact surfaces which is essential for excess pore water pressure generation and therefore crucial for liquefaction [36].

7.1. Implementation phase

In this study, the neural network software (NeuroShell 2) developed by the ward systems group, in US was used for training, testing and validating the GRNN model [38]. This software is capable to implement different neural network architectures including BPNN, GRNN, and PNN. Dataset was randomly categorized into three subsets namely training, testing, and validation sets. Training set generates the algorithm with best smoothing parameter. Testing set is used for observing the capabilities of the generated

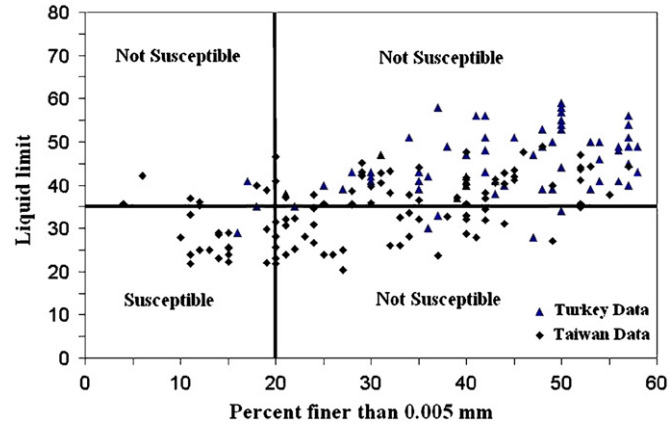

Fig. 5. The distribution of liquid limit vs. percent finer than 0.005 mm for both Turkey and Taiwan earthquake data.

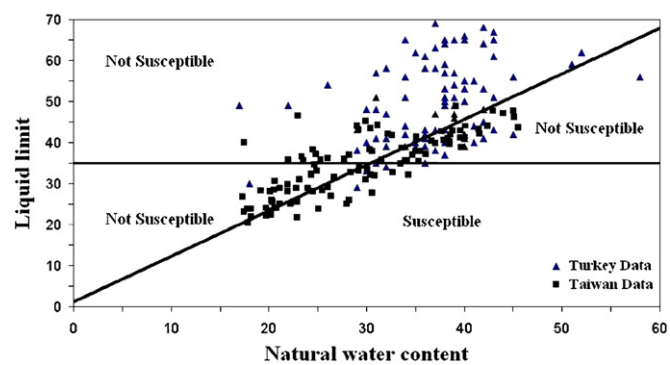

Fig. 6. The distribution of liquid limit vs. natural water content for both Turkey and Taiwan earthquake data.

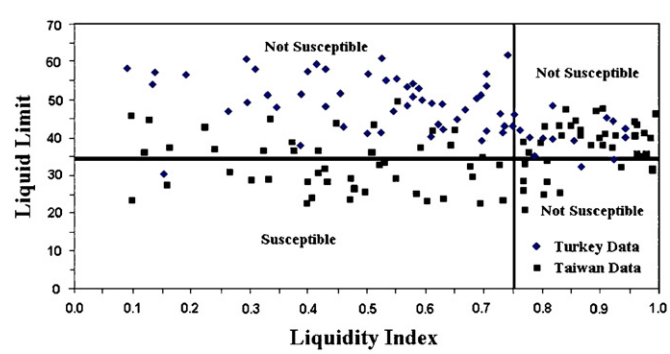

Fig. 7. The distribution of liquid limit vs. liquidity index for both Turkey and Taiwan earthquake data.

algorithm to assess intricate relationships amid input and output values. Validation set was used for applying the trained algorithm to separate dataset, which was not introduced to the network before. Therefore, validation set can be regarded as the real test for the performance of the model.

In this investigation, 413 of these case records were randomly selected for the training phase, 112 for the testing phase, and the remaining 95 records were selected for forecasting. The proposed GRNN model consisted of 12 neurons in the input layer corresponding to the total number of input parameters. The output layer consisted of one neuron for liquefaction decision. The liquefaction potential is characterized by one output neuron; where the binary number 1 represents the occurrence of liquefaction and 0 represents the nonoccurrence of liquefaction.

In the proposed GRNN model, smoothing factor of $\sigma = 0.02552$ was used for soil liquefaction potential estimation. The error limit for the analysis was taken as $\pm 30\%$. This means that when the difference between the target output and the network result within this range, the result is considered to be incorrect. Fig. 8 presents the results of the sensitivity study for the parameters impacting liquefaction potential.

7.2. Validation phase

In this phase, model accuracy and efficiency were examined by making prediction against case records, which were not used during model training and testing. Validation data approximately corresponds to 15% of the dataset selected from Kocaeli, Turkey and Chi-Chi, Taiwan earthquakes case histories. In this phase, the proposed algorithm does not require human development of the proposed model; it rather confirms the architecture prediction capabilities of the model. The results of this phase are also given in Table 2 and summarized in Table 3. Network's outcome in favor of soil liquefaction occurrence/nonoccurrence for both earthquake cases is given in Fig. 9.

8. Conclusions

An AI computational approach, namely GRNN was developed to predict the liquefaction condition in soil deposits. The impact of 12 soil and seismic parameters to soil liquefaction potential in soil deposits has been investigated. In this investigation, a total of 525 case histories were used for the learning and 95 case histories

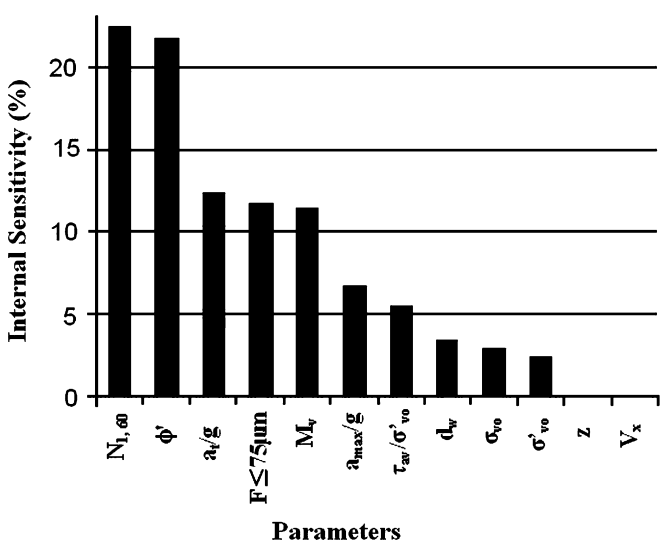

Fig. 8. Parameter sensitivity study.

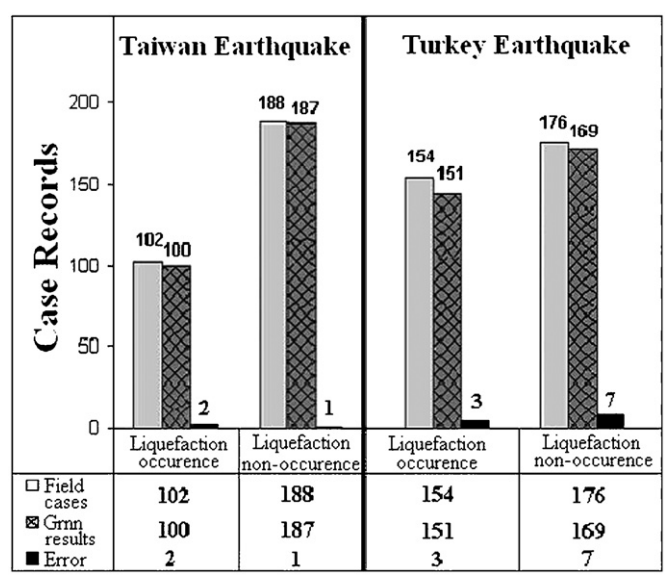

Fig. 9. Network's performance on soil liquefaction potential.

Table 3
Results of the GRNN model

	Overall			Taiwan earthquake			Turkey earthquake		
	Case records	Error	Performance (%)	Case records	Error	Performance (%)	Case records	Error	Performance (%)
Train	413	0	100	193	0	100	220	0	100
Test	112	8	92.9	53	3	94.3	59	5	92
Forecast	95	5	94.7	44	0	100	51	5	90
Total	620	13	97.9	290	3	99.0	330	10	97

were used to make future predictions. The proposed model utilizes reliable soil test results, which were obtained following the two crucial earthquakes in Turkey and Taiwan regions as case records.

A sensitivity analysis was conducted to determine the parameters, which govern this behavior. 12 soil and seismic parameters, which characterizing soil type and material properties, seismic characteristics, magnitude and nature of loads, and other site conditions including stress, strain, strength, saturation and seismological aspects were selected and incorporated into the network. In the parameter sensitivity analysis, it is observed that SPT penetration resistance is the most influential parameter, and shear wave velocity is the least significant parameter. Results are highly consistent with the literature knowledge as it is known that shear wave velocity is a very small-strain measurement and correlates poorly with a much larger strain problem such as liquefaction [32].

The results produced by the proposed "GRNN" model compared well with the determined liquefaction decision obtained by simplified methods. It provides a viable liquefaction potential assessment tool that assist geotechnical engineers in making an accurate and realistic predictions. Furthermore, this study integrates knowledge learned from the two devastating earthquakes to the ongoing development of soil liquefaction analysis.

Acknowledgments

The financial support from the Natural Science and Engineering Research Council of Canada (NSERC) and Concordia University are acknowledged.

Appendix A

For SPT liquefaction case records see Table A1.

Table A1
SPT Liquefaction case records

Z (m)	$(N_1)_{60}$ (—)	$F \leqslant 75\,\mu m$ (%)	d_w (m)	σ_{vo} (kPa)	σ'_{vo} (kPa)	a_t (g)	τ_{av}/σ'_{vo} (—)	V_s (m/s)	ϕ' (°)	M_v (—)	a_{max} (g)	Liquefaction (—)	GRNN (—)
1.0	6	90	0.77	16.3	14.0	0.14	0.29	105	28.48	7.4	0.40	No	No
1.8	8	94	0.77	30.9	20.6	0.08	0.37	105	29.17	7.4	0.40	No	No
2.6	7	100	0.77	45.6	27.3	0.04	0.41	85	26.96	7.4	0.40	No	No
3.4	5	87	0.77	60.3	34.0	0.03	0.43	85	29.68	7.4	0.40	No	No
4.2	5	74	0.77	75.8	41.5	0.02	0.44	85	26.93	7.4	0.40	Yes	Yes
5.0	3	92	0.77	90.1	47.8	0.02	0.45	85	28.39	7.4	0.40	No	No
6.0	3	97	0.77	108.2	55.9	0.15	0.45	250	26.00	7.4	0.40	No	No
7.0	19	70	0.77	127.9	65.6	0.13	0.45	250	33.42	7.4	0.40	Yes	Yes
8.0	26	58	0.77	147.5	75.2	0.11	0.44	250	35.10	7.4	0.40	No	No
9.0	48	5	0.77	167.3	85.0	0.10	0.44	250	42.72	7.4	0.40	No	No
1.0	3	74	0.78	18.0	15.8	0.14	0.29	105	26.00	7.4	0.40	Yes	Yes
1.8	5	86	0.78	32.8	22.6	0.08	0.36	105	26.93	7.4	0.40	Yes	Yes
3.4	2	85	0.78	62.4	36.2	0.03	0.42	85	29.14	7.4	0.40	No	No
4.2	10	93	0.78	77.4	43.2	0.02	0.43	85	29.17	7.4	0.40	Yes	Yes
6.0	4	99	0.78	109.5	57.3	0.15	0.45	250	29.97	7.4	0.40	No	No
7.0	11	85	0.78	128.3	66.1	0.13	0.44	250	30.95	7.4	0.40	Yes	Yes
8.5	39	8	0.78	156.2	79.0	0.11	0.44	250	40.78	7.4	0.40	No	No
10.0	25	6	0.78	186.6	94.4	0.09	0.43	250	37.32	7.4	0.40	No	No
1.8	7	99	0.71	30.3	19.4	0.09	0.39	111	27.34	7.4	0.40	No	No
2.8	4	99	0.71	48.8	27.9	0.07	0.43	121	27.34	7.4	0.40	No	No
3.7	6	79	0.71	66.0	36.1	0.05	0.44	118	27.75	7.4	0.40	Yes	Yes
5.6	5	96	0.71	99.9	51.0	0.03	0.46	105	26.93	7.4	0.40	Yes	Yes
6.5	17	88	0.71	117.0	59.1	0.07	0.46	173	32.96	7.4	0.40	Yes	Yes
8.5	49	9	0.71	157.1	79.2	0.05	0.44	173	42.45	7.4	0.40	No	No
4.1	7	97	0.82	74.1	41.3	0.02	0.43	85	28.49	7.4	0.40	Yes	Yes
5.0	3	98	0.82	90.5	48.7	0.17	0.44	250	29.97	7.4	0.40	No	No
6.6	18	92	0.82	120.0	62.2	0.13	0.45	250	32.99	7.4	0.40	No	No
8.0	32	66	0.82	147.9	76.1	0.11	0.44	250	38.65	7.4	0.40	No	No
9.5	75	8	0.82	179.3	92.5	0.10	0.43	250	44.80	7.4	0.40	No	No
11.0	33	10	0.82	211.1	109.3	0.35	0.41	500	46.35	7.4	0.40	No	No
12.5	33	7	0.82	243.2	126.4	0.31	0.40	500	40.49	7.4	0.40	No	No
15.0	10	100	0.82	289.7	147.9	0.27	0.39	500	25.37	7.4	0.40	No	No
1.6	3	96	3.10	25.3	25.1	0.11	0.25	115	24.90	7.4	0.40	No	No
2.6	3	82	3.10	43.0	42.7	0.07	0.25	115	26.00	7.4	0.40	No	No
3.5	10	21	3.10	58.6	54.6	0.13	0.26	185	29.80	7.4	0.40	Yes	Yes
4.2	12	14	3.10	72.3	61.3	0.11	0.28	185	30.39	7.4	0.40	Yes	Yes
5.0	31	29	3.10	89.3	70.3	0.09	0.30	185	37.66	7.4	0.40	No	No

Table A1 (*continued*)

Z (m)	$(N_1)_{60}$ (—)	$F \leqslant 75\,\mu m$ (%)	d_w (m)	σ_{vo} (kPa)	σ'_{vo} (kPa)	a_t (g)	τ_{av}/σ'_{vo} (—)	V_s (m/s)	ϕ' (°)	M_v (—)	a_{max} (g)	Liquefaction (—)	GRNN (—)
6.2	33	5	3.10	116.4	85.4	0.08	0.32	185	38.65	7.4	0.40	No	No
8.0	31	5	3.10	155.2	106.2	0.29	0.33	400	39.59	7.4	0.40	No	No
10.5	8	100	3.10	202.3	128.3	0.23	0.34	400	26.11	7.4	0.40	No	No
12.4	12	100	3.10	238.1	145.1	0.20	0.34	400	26.11	7.4	0.40	No	No
13.7	9	36	3.10	263.3	157.3	0.19	0.34	400	30.95	7.4	0.40	Yes	Yes
16.0	9	100	3.10	306.1	177.1	0.17	0.34	400	27.15	7.4	0.40	No	No
1.4	4	87	1.68	21.5	21.5	0.12	0.25	115	31.30	7.4	0.40	No	No
3.0	2	91	1.68	50.8	37.6	0.15	0.33	185	24.90	7.4	0.40	Yes	Yes
3.9	15	5	1.68	70.0	47.8	0.12	0.35	185	31.49	7.4	0.40	Yes	Yes
4.7	25	5	1.68	86.4	56.2	0.10	0.37	185	35.09	7.4	0.40	No	No
7.0	31	3	1.68	138.2	85.0	0.33	0.37	400	43.97	7.4	0.40	No	No
8.0	35	4	1.68	159.8	96.6	0.29	0.37	400	44.29	7.4	0.40	No	No
1.8	3	99	1.50	28.7	25.7	0.05	0.28	85	24.90	7.4	0.40	No	No
2.6	4	99	1.50	43.2	32.2	0.04	0.33	85	25.80	7.4	0.40	No	No
4.2	11	98	1.50	72.9	45.9	0.02	0.38	85	29.80	7.4	0.40	Yes	Yes
5.0	18	67	1.50	88.4	53.4	0.09	0.39	180	32.96	7.4	0.40	Yes	Yes
6.0	48	53	1.50	107.8	62.8	0.08	0.40	180	42.45	7.4	0.40	No	No
7.4	20	1	1.50	138.1	79.1	0.06	0.40	180	35.49	7.4	0.40	No	No
9.6	13	89	1.50	180.3	99.3	0.05	0.40	180	31.48	7.4	0.40	Yes	Yes
1.7	3	97	1.34	28.9	25.3	0.16	0.28	145	24.90	7.4	0.40	Yes	Yes
3.3	9	99	1.34	58.0	38.4	0.08	0.37	141	25.37	7.4	0.40	No	No
4.9	10	87	1.34	88.2	52.6	0.02	0.40	85	29.17	7.4	0.40	No	No
5.7	47	71	1.34	104.0	60.4	0.08	0.40	180	40.20	7.4	0.40	No	No
6.5	16	92	1.34	119.5	67.9	0.07	0.41	180	31.49	7.4	0.40	Yes	Yes
7.6	7	99	1.34	139.7	77.1	0.06	0.41	180	28.88	7.4	0.40	No	No
9.0	9	99	1.34	165.6	89.0	0.05	0.41	180	29.17	7.4	0.40	Yes	Yes
3.8	10	28	1.30	68.2	43.2	0.04	0.38	100	29.17	7.4	0.40	Yes	Yes
4.6	4	88	1.30	82.8	49.8	0.07	0.40	155	26.00	7.4	0.40	Yes	Yes
5.5	24	97	1.30	99.7	57.7	0.06	0.41	155	30.60	7.4	0.40	No	No
6.7	12	83	1.30	122.5	68.5	0.05	0.41	155	31.49	7.4	0.40	Yes	Yes
7.7	6	98	1.30	140.5	76.5	0.04	0.42	155	27.75	7.4	0.40	No	No
9.8	23	75	1.30	182.3	97.3	0.04	0.41	155	35.87	7.4	0.40	Yes	Yes
1.4	4	99	0.44	23.9	14.3	0.09	0.42	100	29.97	7.4	0.40	No	No
2.4	2	99	0.44	41.8	22.2	0.06	0.47	100	29.97	7.4	0.40	No	No
3.5	3	99	0.44	61.5	30.9	0.04	0.48	100	29.97	7.4	0.40	No	No
4.4	7	51	0.44	78.9	39.3	0.07	0.48	155	27.75	7.4	0.40	Yes	Yes
3.0	1	96	0.44	51.8	26.7	0.03	0.47	85	29.40	7.4	0.40	No	No
5.1	5	100	0.44	90.8	44.2	0.09	0.49	180	31.30	7.4	0.40	No	No
6.0	55	99	0.44	107.3	51.7	0.08	0.48	180	41.91	7.4	0.40	No	No
7.0	44	4	0.44	129.0	63.4	0.07	0.47	180	40.20	7.4	0.40	No	No
1.0	3	94	0.96	15.1	14.7	0.09	0.26	85	24.90	7.4	0.40	Yes	No[a]
1.8	1	87	0.96	29.9	21.5	0.05	0.35	85	23.46	7.4	0.40	Yes	Yes
2.5	16	99	0.96	42.3	26.9	0.04	0.39	85	27.34	7.4	0.40	No	No
1.0	6	84	0.96	15.1	14.7	0.09	0.26	85	26.93	7.4	0.40	Yes	Yes
1.9	4	72	0.96	31.1	22.2	0.05	0.35	85	26.00	7.4	0.40	Yes	Yes
2.6	4	99	0.96	47.3	30.9	0.04	0.38	85	28.39	7.4	0.40	No	No
3.3	7	99	0.96	60.3	36.9	0.03	0.40	85	25.65	7.4	0.40	No	No
1.1	1	81	1.69	21.6	21.6	0.20	0.25	130	32.99	7.4	0.40	No	No
1.9	2	71	1.69	36.1	34.0	0.15	0.26	145	24.90	7.4	0.40	Yes	Yes
2.7	2	99	1.69	48.0	37.9	0.10	0.31	145	28.88	7.4	0.40	No	No
3.4	9	89	1.69	61.0	43.9	0.08	0.34	145	34.16	7.4	0.40	Yes	Yes
4.4	30	4	1.69	82.5	55.4	0.07	0.36	145	41.61	7.4	0.40	No	No
5.5	30	3	1.69	107.3	69.2	0.05	0.36	145	41.97	7.4	0.40	No	No
7.5	36	5	1.69	147.2	89.1	0.04	0.38	145	44.60	7.4	0.40	No	No
10.0	8	94	1.69	194.3	111.2	0.04	0.38	170	29.17	7.4	0.40	No	No
2.6	3	99	1.90	42.6	35.6	0.33	0.29	256	26.00	7.4	0.40	No	No
3.3	3	95	1.90	55.6	41.6	0.27	0.33	256	26.00	7.4	0.40	No	No
4.1	25	71	1.90	70.0	48.0	0.31	0.35	306	34.70	7.4	0.40	No	No
5.2	30	4	1.90	95.0	62.0	0.17	0.36	251	41.97	7.4	0.40	No	No
7.0	31	5	1.90	136.1	85.1	0.13	0.37	251	43.97	7.4	0.40	No	No
3.2	5	52	2.28	52.1	42.9	0.09	0.30	145	26.93	7.4	0.40	Yes	Yes
3.8	11	55	2.28	63.8	48.6	0.07	0.32	145	30.95	7.4	0.40	Yes	Yes
4.6	22	4	2.28	80.6	57.4	0.06	0.33	145	40.10	7.4	0.40	Yes	Yes

Table A1 (*continued*)

Z (m)	$(N_1)_{60}$ (—)	$F \leqslant 75\,\mu m$ (%)	d_w (m)	σ_{vo} (kPa)	σ'_{vo} (kPa)	a_t (g)	τ_{av}/σ'_{vo} (—)	V_s (m/s)	ϕ' (°)	M_v (—)	a_{max} (g)	Liquefaction (—)	GRNN (—)
1.6	10	2	0.50	28.5	17.5	0.03	0.41	59	28.48	7.4	0.40	Yes	Yes
2.9	7	59	0.50	52.9	28.9	0.02	0.45	67	27.75	7.4	0.40	Yes	Yes
3.8	4	99	0.50	68.5	35.5	0.02	0.47	67	26.00	7.4	0.40	No	No
5.4	11	47	0.50	98.4	49.4	0.04	0.47	119	30.39	7.4	0.40	Yes	Yes
6.4	38	61	0.50	118.4	59.4	0.03	0.46	119	39.59	7.4	0.40	No	No
8.8	13	3	0.50	167.1	84.1	0.15	0.44	299	32.00	7.4	0.40	Yes	Yes
2.2	9	5	0.44	41.2	23.6	0.02	0.43	59	28.49	7.4	0.40	Yes	Yes
1.6	18	5	0.35	28.9	16.4	0.04	0.44	72	32.00	7.4	0.40	Yes	Yes
2.4	4	2	0.35	46.4	25.9	0.02	0.44	59	26.00	7.4	0.40	Yes	Yes
3.2	9	59	0.35	61.3	32.8	0.02	0.46	67	28.49	7.4	0.40	Yes	Yes
4.0	4	99	0.35	75.1	38.6	0.02	0.47	67	27.14	7.4	0.40	No	No
5.4	24	98	0.35	101.0	50.5	0.01	0.47	60	29.97	7.4	0.40	No	No
1.0	9	99	0.74	15.8	13.2	0.14	0.30	105	24.96	7.4	0.40	No	No
2.9	4	64	0.74	55.3	33.7	0.03	0.40	85	26.00	7.4	0.40	Yes	Yes
4.0	5	89	0.74	75.6	43.0	0.02	0.42	85	26.93	7.4	0.40	No	No[a]
4.8	12	94	0.74	90.0	49.4	0.02	0.43	85	27.34	7.4	0.40	No	No
6.1	12	96	0.74	114.0	60.4	0.14	0.44	250	30.95	7.4	0.40	Yes	Yes
7.0	31	8	0.74	132.1	69.5	0.13	0.44	250	38.33	7.4	0.40	No	No
7.9	44	9	0.74	150.4	78.8	0.11	0.43	250	42.18	7.4	0.40	No	No
8.8	44	5	0.74	169.1	88.5	0.10	0.42	250	43.00	7.4	0.40	No	No
1.1	4	99	0.80	17.9	14.9	0.06	0.30	72	29.68	7.4	0.40	No	No
3.0	6	85	0.80	53.2	31.2	0.02	0.42	67	27.75	7.4	0.40	Yes	Yes
3.8	4	95	0.80	68.0	38.0	0.02	0.43	67	26.00	7.4	0.40	No	No
4.6	4	98	0.80	82.6	44.6	0.01	0.44	59	30.28	7.4	0.40	No	No
5.5	24	89	0.80	99.5	52.5	0.01	0.45	59	35.49	7.4	0.40	Yes	Yes
6.5	38	8	0.80	119.6	62.6	0.03	0.44	117	38.97	7.4	0.40	No	No
8.0	18	25	0.80	148.1	76.1	0.16	0.44	299	32.96	7.4	0.40	Yes	Yes
9.1	15	81	0.80	169.3	86.3	0.14	0.43	299	32.49	7.4	0.40	Yes	Yes
1.2	3	97	0.60	20.0	14.0	0.13	0.36	110	24.90	7.4	0.40	No	No
2.0	7	93	0.60	34.7	20.7	0.08	0.42	110	27.75	7.4	0.40	Yes	Yes
2.8	9	92	0.60	49.6	27.6	0.06	0.44	110	28.49	7.4	0.40	Yes	Yes
3.6	3	96	0.60	64.2	34.2	0.03	0.45	90	28.17	7.4	0.40	No	No
4.4	5	94	0.60	78.6	40.6	0.03	0.46	90	26.93	7.4	0.40	No	No
5.2	18	18	0.60	94.5	48.5	0.05	0.46	135	32.49	7.4	0.40	Yes	Yes
6.0	36	88	0.60	110.6	56.6	0.04	0.46	135	33.33	7.4	0.40	No	No
6.8	15	14	0.60	126.1	64.1	0.04	0.45	135	32.00	7.4	0.40	Yes	Yes
8.3	10	91	0.60	154.7	77.7	0.03	0.45	135	29.80	7.4	0.40	Yes	Yes
10.2	27	19	0.60	193.0	97.0	0.03	0.43	135	38.00	7.4	0.40	No	No
1.8	4	81	0.84	30.6	21.0	0.09	0.36	110	26.00	7.4	0.40	Yes	Yes
2.7	12	53	0.84	47.5	28.9	0.06	0.40	110	29.80	7.4	0.40	Yes	Yes
4.0	4	99	0.84	70.3	38.7	0.03	0.44	90	25.51	7.4	0.40	No	No
5.2	20	39	0.84	93.9	50.3	0.05	0.44	135	33.42	7.4	0.40	Yes	Yes
6.3	21	99	0.84	114.3	59.7	0.04	0.44	135	34.28	7.4	0.40	Yes	Yes
7.1	13	98	0.84	129.5	66.9	0.04	0.44	135	31.49	7.4	0.40	Yes	No[a]
8.4	11	100	0.84	153.2	77.6	0.03	0.44	135	26.11	7.4	0.40	No	No
3.5	8	98	0.75	63.1	35.6	0.03	0.43	90	28.49	7.4	0.40	Yes	Yes
5.5	7	72	0.75	101.0	53.5	0.05	0.44	135	28.49	7.4	0.40	Yes	Yes
6.5	36	64	0.75	119.0	61.5	0.04	0.45	135	34.28	7.4	0.40	No	No
9.0	7	97	0.75	165.1	82.6	0.03	0.44	135	27.34	7.4	0.40	No	No
1.2	3	91	0.40	20.6	12.6	0.13	0.41	110	26.00	7.4	0.40	Yes	Yes
2.0	6	56	0.40	36.1	20.1	0.08	0.45	110	27.75	7.4	0.40	Yes	Yes
2.8	8	65	0.40	50.7	26.7	0.06	0.47	110	30.95	7.4	0.40	Yes	Yes
3.8	5	97	0.40	69.1	35.1	0.03	0.48	90	28.88	7.4	0.40	No	No
4.4	7	99	0.40	79.9	39.9	0.03	0.48	90	27.75	7.4	0.40	Yes	Yes
1.2	3	75	1.72	20.0	20.0	0.03	0.25	49	29.68	7.4	0.40	No	No
2.2	15	15	1.72	39.9	35.1	0.01	0.28	49	31.49	7.4	0.40	Yes	Yes
3.0	3	98	1.72	54.1	41.3	0.02	0.32	66	26.00	7.4	0.40	No	No
3.8	3	100	1.72	68.2	47.4	0.02	0.35	66	26.00	7.4	0.40	No	No
4.9	5	99	1.72	88.8	57.0	0.03	0.37	96	31.68	7.4	0.40	No	No
5.7	14	80	1.72	104.0	64.2	0.02	0.38	95	32.49	7.4	0.40	Yes	Yes
6.5	8	99	1.72	118.6	70.8	0.02	0.39	95	29.17	7.4	0.40	Yes	Yes
7.3	9	99	1.72	133.2	77.4	0.01	0.39	81	29.80	7.4	0.40	No	Yes[a]
8.5	7	98	1.72	154.6	86.8	0.01	0.40	67	29.17	7.4	0.40	Yes	Yes

Table A1 (*continued*)

Z (m)	$(N_1)_{60}$ (—)	$F \leqslant 75\,\mu m$ (%)	d_w (m)	σ_{vo} (kPa)	σ'_{vo} (kPa)	a_t (g)	τ_{av}/σ'_{vo} (—)	V_s (m/s)	ϕ' (°)	M_v (—)	a_{max} (g)	Liquefaction (—)	GRNN (—)
9.9	9	9	1.72	181.7	99.9	0.01	0.40	65	30.39	7.4	0.40	Yes	Yes
11.0	7	98	1.72	202.0	109.2	0.01	0.40	67	25.23	7.4	0.40	No	No
1.0	7	72	1.64	15.0	15.0	0.83	0.25	253	27.75	7.4	0.40	No	No
1.8	5	68	1.64	30.6	29.0	0.47	0.26	253	26.93	7.4	0.40	Yes	Yes
2.6	6	92	1.64	45.6	36.0	0.33	0.31	253	27.75	7.4	0.40	Yes	Yes
3.6	3	97	1.64	64.0	44.4	0.09	0.35	151	26.00	7.4	0.40	No	No
4.5	12	85	1.64	81.2	52.6	0.01	0.37	44	33.48	7.4	0.40	Yes	Yes
5.4	18	51	1.64	98.9	61.3	0.01	0.38	47	32.96	7.4	0.40	Yes	Yes
6.2	16	72	1.64	113.8	68.2	0.01	0.39	49	32.96	7.4	0.40	Yes	Yes
7.0	14	42	1.64	129.6	76.0	0.00	0.39	49	32.49	7.4	0.40	Yes	Yes
8.0	13	74	1.64	148.9	85.3	0.02	0.39	95	32.49	7.4	0.40	Yes	Yes
9.0	11	100	1.64	167.4	93.8	0.01	0.40	95	28.17	7.4	0.40	No	No[a]
10.0	10	96	1.64	186.2	102.6	0.01	0.39	92	30.39	7.4	0.40	No	No
12.0	27	90	1.64	225.0	121.4	0.01	0.39	88	34.89	7.4	0.40	No	No
1.5	4	97	0.54	25.4	15.8	0.01	0.40	40	26.00	7.4	0.40	No	No
2.3	9	69	0.54	40.5	22.9	0.15	0.44	161	29.17	7.4	0.40	Yes	Yes
3.1	5	95	0.54	55.2	29.6	0.01	0.45	51	26.93	7.4	0.40	Yes	Yes
4.0	5	97	0.54	71.9	37.3	0.00	0.46	37	27.34	7.4	0.40	No	No
4.8	13	84	0.54	87.5	44.9	0.00	0.46	37	29.14	7.4	0.40	No	No
5.6	10	90	0.54	102.4	51.8	0.03	0.46	102	29.80	7.4	0.40	Yes	Yes
7.0	12	67	0.54	129.9	65.3	0.04	0.46	134	30.95	7.4	0.40	Yes	Yes
8.0	7	99	0.54	148.4	73.8	0.03	0.45	134	28.48	7.4	0.40	Yes	No[a]
9.0	9	99	0.54	167.2	82.6	0.01	0.45	65	29.17	7.4	0.40	No	Yes[a]
2.5	13	65	0.45	42.4	21.9	0.06	0.48	105	30.39	7.4	0.40	Yes	Yes
5.2	8	75	0.45	92.8	45.3	0.04	0.48	120	28.48	7.4	0.40	Yes	Yes
6.0	18	99	0.45	107.5	52.0	0.05	0.48	145	32.49	7.4	0.40	Yes	Yes
8.5	9	89	0.45	153.6	73.1	0.04	0.47	145	29.80	7.4	0.40	Yes	Yes
11.0	9	98	0.45	201.9	96.4	0.03	0.45	145	27.95	7.4	0.40	No	No
12.0	28	97	0.45	220.3	104.8	0.03	0.44	145	37.32	7.4	0.40	No	No
14.0	12	99	0.45	258.2	122.7	0.02	0.43	145	31.30	7.4	0.40	No	No
2.0	30	11	10.00	37.3	37.3	0.14	0.25	146	41.24	7.4	0.40	No	No
3.5	14	34	10.00	66.2	66.2	0.08	0.24	145	32.00	7.4	0.40	No	No
5.0	13	82	10.00	91.2	91.2	0.06	0.24	145	26.78	7.4	0.40	No	No
6.6	8	35	10.00	122.0	122.0	0.05	0.23	145	29.80	7.4	0.40	No	No
8.0	48	89	10.00	146.7	146.7	0.05	0.23	170	34.05	7.4	0.40	No	No
2.3	5	65	1.50	38.3	30.3	0.06	0.31	100	26.93	7.4	0.40	Yes	Yes
3.4	4	78	1.50	56.8	38.3	0.09	0.36	150	31.30	7.4	0.40	No	No
4.1	9	33	1.50	71.2	45.2	0.07	0.38	150	28.49	7.4	0.40	Yes	Yes
5.0	6	99	1.50	84.6	50.1	0.06	0.40	150	26.96	7.4	0.40	No	No
5.8	10	98	1.50	99.6	56.6	0.05	0.41	150	29.80	7.4	0.40	Yes	Yes
6.8	43	17	1.50	117.9	65.4	0.05	0.41	150	40.19	7.4	0.40	No	No
7.9	40	11	1.50	138.8	74.8	0.04	0.42	150	39.90	7.4	0.40	No	No
9.0	50	4	1.50	157.0	82.5	0.04	0.42	150	42.71	7.4	0.40	No	No
0.9	7	97	0.95	13.7	13.7	0.16	0.25	105	28.40	7.4	0.40	No	No
1.7	6	89	0.95	27.8	20.3	0.08	0.34	105	26.93	7.4	0.40	Yes	Yes
2.6	6	89	0.95	43.7	27.2	0.04	0.40	85	26.93	7.4	0.40	Yes	Yes
3.7	4	100	0.95	61.0	33.5	0.03	0.44	85	28.40	7.4	0.40	No	No
4.8	4	97	0.95	79.5	41.5	0.02	0.46	85	26.00	7.4	0.40	Yes	Yes
5.8	5	99	0.95	97.2	49.2	0.02	0.46	85	26.93	7.4	0.40	No	No
6.7	17	54	0.95	113.9	56.4	0.13	0.46	250	32.96	7.4	0.40	Yes	Yes
8.0	6	100	0.95	134.4	63.9	0.11	0.47	250	27.34	7.4	0.40	No	No
9.1	9	99	0.95	151.6	70.1	0.10	0.48	250	26.44	7.4	0.40	No	No
5.0	1	86	0.90	85.2	44.7	0.04	0.45	120	27.54	7.4	0.40	No	No
7.0	1	96	0.90	117.4	56.4	0.03	0.48	120	27.54	7.4	0.40	No	No
4.4	4	97	1.10	82.7	49.7	0.03	0.40	95	30.94	7.4	0.40	No	No
6.3	5	98	1.10	116.3	64.3	0.02	0.42	95	31.30	7.4	0.40	No	No
7.4	4	98	1.10	134.8	72.3	0.02	0.42	95	31.30	7.4	0.40	No	No
10.5	10	78	1.10	190.5	96.5	0.03	0.43	135	26.61	7.4	0.40	No	No
3.8	1	93	1.20	65.7	39.7	0.02	0.40	70	31.68	7.4	0.40	No	No
7.5	13	100	1.20	131.0	68.0	0.01	0.44	70	30.94	7.4	0.40	No	No
1.0	8	98	1.11	15.0	15.0	0.17	0.25	115	26.27	7.4	0.40	No	No
2.2	9	98	1.11	38.1	27.2	0.08	0.35	115	26.96	7.4	0.40	No	No
3.5	9	71	1.11	62.9	39.0	0.05	0.39	115	29.17	7.4	0.40	Yes	Yes

Table A1 (*continued*)

Z (m)	$(N_1)_{60}$ (—)	$F \leqslant 75\,\mu m$ (%)	d_w (m)	σ_{vo} (kPa)	σ'_{vo} (kPa)	a_t (g)	τ_{av}/σ'_{vo} (—)	V_s (m/s)	ϕ' (°)	M_v (—)	a_{max} (g)	Liquefaction (—)	GRNN (—)
4.5	8	97	1.11	81.1	47.2	0.04	0.41	115	24.58	7.4	0.40	No	No
5.5	17	99	1.11	100.4	56.5	0.05	0.42	140	24.70	7.4	0.40	No	No
7.0	9	100	1.11	128.8	69.9	0.04	0.42	140	25.65	7.4	0.40	No	No
8.0	3	99	1.11	147.5	78.6	0.04	0.42	140	29.14	7.4	0.40	No	No[a]
9.0	5	95	1.11	166.5	87.6	0.04	0.42	160	27.75	7.4	0.40	Yes	Yes
1.2	12	99	2.40	18.2	18.2	0.14	0.25	115	26.27	7.4	0.40	No	No
2.2	4	77	2.40	35.9	35.9	0.08	0.25	115	26.93	7.4	0.40	No	No
3.0	7	94	2.40	51.2	45.2	0.06	0.28	115	28.63	7.4	0.40	No	No[a]
3.8	7	68	2.40	66.2	52.2	0.05	0.31	115	27.75	7.4	0.40	Yes	Yes
5.0	5	94	2.40	88.7	62.7	0.05	0.33	140	30.28	7.4	0.40	No	No
5.8	10	99	2.40	103.6	69.6	0.05	0.35	140	29.80	7.4	0.40	Yes	Yes
6.6	7	20	2.40	119.3	77.3	0.04	0.36	140	28.49	7.4	0.40	Yes	Yes
7.6	7	80	2.40	141.6	89.6	0.04	0.36	140	28.49	7.4	0.40	Yes	Yes
8.5	9	78	2.40	159.0	98.0	0.03	0.36	140	29.80	7.4	0.40	Yes	Yes
9.5	4	78	2.40	178.1	107.1	0.04	0.37	160	26.93	7.4	0.40	Yes	Yes
1.9	14	82	0.37	22.8	7.5	0.07	0.76	100	29.17	7.4	0.40	Yes	Yes
2.8	5	98	0.37	38.9	14.6	0.05	0.65	100	28.88	7.4	0.40	No	No
3.8	6	82	0.37	57.7	23.4	0.08	0.60	150	26.93	7.4	0.40	Yes	Yes
5.0	21	65	0.37	80.6	34.3	0.06	0.56	150	33.42	7.4	0.40	Yes	Yes
6.2	15	81	0.37	103.8	45.5	0.05	0.53	150	29.97	7.4	0.40	Yes	Yes
7.2	21	13	0.37	123.3	55.0	0.04	0.51	150	34.28	7.4	0.40	Yes	Yes
8.4	31	7	0.37	148.0	67.7	0.04	0.49	150	38.00	7.4	0.40	No	No
9.2	7	99	0.37	162.2	73.9	0.04	0.48	160	25.51	7.4	0.40	No	No
2.8	6	65	2.60	26.8	24.9	0.06	0.25	100	26.93	7.4	0.40	Yes	Yes
3.3	4	78	2.60	47.9	40.9	0.09	0.28	150	31.30	7.4	0.40	No	No
4.0	8	33	2.60	61.4	47.4	0.08	0.31	150	28.49	7.4	0.40	Yes	Yes
4.9	6	99	2.60	77.6	54.6	0.06	0.34	150	26.96	7.4	0.40	No	No
5.7	10	98	2.60	92.2	61.2	0.06	0.35	150	29.80	7.4	0.40	Yes	Yes
6.7	42	17	2.60	112.7	71.7	0.05	0.36	150	40.20	7.4	0.40	No	No
7.7	39	11	2.60	133.0	82.0	0.04	0.37	150	39.90	7.4	0.40	No	No
8.9	47	4	2.60	160.1	97.1	0.04	0.37	150	42.72	7.4	0.40	No	No
1.4	2	51	0.74	23.4	16.8	0.16	0.35	130	23.46	7.4	0.40	Yes	Yes
2.5	6	83	0.74	43.9	26.3	0.09	0.41	130	26.93	7.4	0.40	Yes	Yes
3.3	6	83	0.74	59.0	33.4	0.12	0.43	170	26.93	7.4	0.40	Yes	Yes
4.2	5	94	0.74	75.4	40.8	0.09	0.44	170	26.93	7.4	0.40	Yes	Yes
5.0	6	92	0.74	91.5	48.9	0.08	0.44	170	27.95	7.4	0.40	No	No
5.7	14	9	0.74	105.5	55.9	0.07	0.44	170	32.00	7.4	0.40	Yes	Yes
6.5	45	5	0.74	122.1	64.5	0.06	0.44	170	41.63	7.4	0.40	No	No
7.5	39	8	0.74	142.7	75.1	0.06	0.43	170	40.78	7.4	0.40	No	No
8.5	7	98	0.74	161.1	83.5	0.06	0.43	190	29.14	7.4	0.40	No	No
1.4	3	51	0.62	23.9	16.1	0.16	0.37	130	24.90	7.4	0.40	Yes	Yes
2.3	7	9	0.62	41.9	25.1	0.10	0.41	135	32.74	7.4	0.40	Yes	Yes
3.2	6	98	0.62	58.7	32.9	0.08	0.43	135	32.99	7.4	0.40	No	No
4.0	4	100	0.62	73.4	39.6	0.06	0.45	135	26.00	7.4	0.40	Yes	Yes
5.0	6	100	0.62	93.0	49.2	0.05	0.45	135	27.75	7.4	0.40	Yes	Yes
6.0	10	98	0.62	111.4	57.6	0.04	0.45	135	31.30	7.4	0.40	No	No
6.8	13	97	0.62	126.1	64.3	0.04	0.45	135	32.09	7.4	0.40	No	No
8.0	9	9	0.62	150.2	76.4	0.03	0.44	135	34.80	7.4	0.40	Yes	Yes
9.2	4	97	0.62	172.9	87.1	0.03	0.44	135	29.68	7.4	0.40	No	No[a]
1.0	7	97	0.92	17.6	16.8	0.13	0.26	100	26.44	7.4	0.40	No	No
2.0	3	87	0.92	36.6	25.8	0.07	0.35	100	24.90	7.4	0.40	Yes	Yes
3.0	7	90	0.92	55.0	34.2	0.10	0.39	150	27.75	7.4	0.40	Yes	Yes
4.0	7	96	0.92	73.3	42.5	0.08	0.42	150	27.75	7.4	0.40	Yes	Yes
5.0	4	100	0.92	91.3	50.5	0.06	0.43	150	25.80	7.4	0.40	No	No
5.8	7	88	0.92	106.4	57.6	0.05	0.43	150	33.50	7.4	0.40	Yes	Yes
6.7	5	98	0.92	122.9	65.1	0.05	0.43	150	28.40	7.4	0.40	Yes	Yes
8.0	7	100	0.92	146.6	75.8	0.04	0.44	150	26.11	7.4	0.40	No	No
9.2	32	20	0.92	170.8	88.0	0.04	0.43	160	39.29	7.4	0.40	No	No
10.0	45	11	0.92	187.1	96.3	0.04	0.42	160	43.24	7.4	0.40	No	No
1.0	5	95	0.77	18.1	8.1	0.13	0.56	100	26.78	7.4	0.40	No	No
1.8	5	99	0.77	32.5	14.5	0.07	0.56	100	25.51	7.4	0.40	No	No
2.9	5	91	0.77	53.0	24.0	0.05	0.54	100	32.09	7.4	0.40	Yes	Yes
4.0	3	78	0.77	73.7	33.7	0.08	0.53	150	24.90	7.4	0.40	Yes	Yes

Table A1 (*continued*)

Z (m)	$(N_1)_{60}$ (—)	$F \leqslant 75\,\mu m$ (%)	d_w (m)	σ_{vo} (kPa)	σ'_{vo} (kPa)	a_t (g)	τ_{av}/σ'_{vo} (—)	V_s (m/s)	ϕ' (°)	M_v (—)	a_{max} (g)	Liquefaction (—)	GRNN (—)
5.0	16	92	0.77	92.9	42.9	0.06	0.51	150	32.00	7.4	0.40	Yes	Yes
6.1	16	80	0.77	114.6	53.6	0.05	0.50	150	32.49	7.4	0.40	Yes	Yes
7.5	38	41	0.77	142.6	67.6	0.04	0.48	150	40.20	7.4	0.40	No	No
8.5	40	20	0.77	162.8	77.8	0.04	0.47	150	41.07	7.4	0.40	No	No
9.4	34	24	0.77	180.9	86.9	0.04	0.46	160	39.90	7.4	0.40	No	No
1.3	5	84	0.50	20.5	12.5	0.10	0.41	100	26.00	7.4	0.40	Yes	Yes
2.1	11	91	0.50	35.7	19.7	0.06	0.45	100	29.17	7.4	0.40	Yes	Yes
3.0	6	97	0.50	51.5	26.5	0.10	0.47	150	26.93	7.4	0.40	Yes	Yes
4.0	7	83	0.50	69.7	34.7	0.08	0.48	150	27.75	7.4	0.40	Yes	Yes
5.0	15	54	0.50	89.3	44.3	0.06	0.48	150	32.00	7.4	0.40	Yes	Yes
6.0	16	99	0.50	108.1	53.1	0.05	0.47	150	32.49	7.4	0.40	Yes	Yes
7.0	13	95	0.50	126.4	61.4	0.05	0.47	150	31.49	7.4	0.40	Yes	Yes
8.0	6	99	0.50	144.7	69.7	0.04	0.47	150	28.63	7.4	0.40	No	No
9.0	8	100	0.50	162.9	77.9	0.04	0.46	160	29.17	7.4	0.40	Yes	Yes
1.0	3	89	1.16	15.5	15.5	0.13	0.25	100	30.60	7.4	0.40	No	No
2.0	7	57	1.16	34.7	26.3	0.07	0.33	100	27.75	7.4	0.40	Yes	Yes
3.0	4	99	1.16	52.5	34.1	0.10	0.38	150	26.11	7.4	0.40	No	No
3.9	4	92	1.16	69.6	42.2	0.08	0.40	150	26.00	7.4	0.40	Yes	Yes
5.0	15	35	1.16	91.4	53.0	0.06	0.41	150	32.00	7.4	0.40	Yes	Yes
6.0	10	82	1.16	110.6	62.2	0.05	0.41	150	30.39	7.4	0.40	Yes	Yes
7.0	11	93	1.16	128.5	70.1	0.05	0.42	150	30.95	7.4	0.40	Yes	Yes
8.0	13	58	1.16	147.7	79.3	0.04	0.42	150	32.00	7.4	0.40	Yes	Yes
9.0	8	99	1.16	166.9	88.5	0.04	0.42	160	28.40	7.4	0.40	Yes	Yes
1.0	7	87	0.64	16.1	12.5	0.13	0.32	100	30.94	7.4	0.40	No	No
2.0	12	43	0.64	34.7	21.1	0.07	0.41	100	29.80	7.4	0.40	Yes	Yes
3.0	3	57	0.64	53.9	30.3	0.10	0.43	150	24.90	7.4	0.40	Yes	Yes
4.0	5	95	0.64	72.3	38.7	0.08	0.45	150	27.54	7.4	0.40	No	No
5.0	11	91	0.64	91.6	48.0	0.06	0.45	150	30.39	7.4	0.40	Yes	Yes
6.0	12	58	0.64	111.2	57.6	0.05	0.45	150	30.95	7.4	0.40	Yes	Yes
7.0	10	93	0.64	130.2	66.6	0.05	0.45	150	30.39	7.4	0.40	Yes	Yes
8.0	9	89	0.64	147.7	74.1	0.04	0.45	150	29.80	7.4	0.40	Yes	Yes
8.8	7	100	0.64	162.2	80.6	0.04	0.45	150	29.17	7.4	0.40	Yes	Yes
9.8	34	7	0.64	183.3	91.7	0.04	0.44	160	39.90	7.4	0.40	No	No
1.1	11	89	1.65	17.7	17.7	0.12	0.25	100	26.96	7.4	0.40	No	No
3.5	5	65	1.65	65.5	47.0	0.09	0.34	150	26.93	7.4	0.40	Yes	Yes
4.7	6	97	1.65	86.2	55.7	0.07	0.37	150	26.96	7.4	0.40	No	No
5.8	31	34	1.65	108.2	66.7	0.05	0.38	150	38.00	7.4	0.40	No	No
7.0	9	84	1.65	132.5	79.0	0.05	0.38	150	29.80	7.4	0.40	Yes	Yes
8.3	10	96	1.65	157.5	91.0	0.04	0.39	150	30.95	7.4	0.40	Yes	Yes
9.2	7	94	1.65	174.1	98.6	0.04	0.39	160	29.17	7.4	0.40	Yes	Yes
10.0	29	32	1.65	190.3	106.8	0.04	0.39	160	38.97	7.4	0.40	No	No
2.8	8	18	0.60	55.2	33.6	0.11	0.19	157	27.75	7.6	0.18	Yes	Yes
4.4	8	98	0.60	88.3	51.0	0.07	0.20	157	31.88	7.6	0.18	No	No
5.8	2	60	0.60	113.6	62.6	0.06	0.21	157	34.28	7.6	0.18	No	No
7.2	10	98	0.60	140.5	75.7	0.05	0.21	157	29.94	7.6	0.18	No	No
8.8	5	98	0.60	168.6	88.2	0.04	0.21	157	32.13	7.6	0.18	No	No
10.2	5	99	0.60	186.9	96.7	0.04	0.21	157	30.54	7.6	0.18	No	No
11.8	4	90	0.60	222.7	112.9	0.03	0.20	157	26.00	7.6	0.18	Yes	Yes
13.2	18	21	0.60	250.5	126.9	0.06	0.19	227	33.86	7.6	0.18	No	No
16.2	13	13	0.60	313.0	160.0	0.05	0.17	227	32.49	7.6	0.18	Yes	Yes
17.2	18	16	0.60	333.5	170.7	0.05	0.16	227	34.70	7.6	0.18	No	No
18.8	18	16	0.60	367.5	189.0	0.05	0.15	227	35.10	7.6	0.18	No	No
1.2	6	76	0.70	18.8	13.9	0.20	0.16	138	33.29	7.6	0.18	No	No
4.6	8	49	0.70	82.5	44.2	0.05	0.21	125	27.75	7.6	0.18	Yes	Yes
5.6	5	99	0.70	101.3	53.2	0.07	0.22	175	33.45	7.6	0.18	No	No
9.6	8	97	0.70	175.6	88.3	0.05	0.22	175	34.05	7.6	0.18	No	No
12.2	12	96	0.70	224.0	111.2	0.04	0.20	175	35.04	7.6	0.18	No	No
13.2	9	80	0.70	243.0	120.4	0.04	0.20	175	34.46	7.6	0.18	No	No
14.8	12	80	0.70	276.0	137.7	0.03	0.18	175	34.71	7.6	0.18	No	No
16.2	12	85	0.70	304.3	152.2	0.06	0.17	250	33.04	7.6	0.18	No	No
17.2	40	47	0.70	324.2	162.4	0.06	0.16	250	41.91	7.6	0.18	No	No
1.2	6	44	2.50	18.7	18.7	0.24	0.12	150	28.15	7.6	0.18	No	No
2.8	6	90	2.50	45.7	42.7	0.10	0.12	150	33.60	7.6	0.18	No	No

Table A1 (*continued*)

Z (m)	$(N_1)_{60}$ (—)	$F \leqslant 75\,\mu m$ (%)	d_w (m)	σ_{vo} (kPa)	σ'_{vo} (kPa)	a_t (g)	τ_{av}/σ'_{vo} (—)	V_s (m/s)	ϕ' (°)	M_v (—)	a_{max} (g)	Liquefaction (—)	GRNN (—)
4.2	6	97	2.50	72.5	55.9	0.07	0.15	150	32.44	7.6	0.18	No	No
5.8	4	33	2.50	103.3	71.0	0.05	0.17	150	26.00	7.6	0.18	Yes	Yes
7.2	9	90	2.50	129.8	83.7	0.06	0.17	172	29.17	7.6	0.18	No	No
8.8	11	99	2.50	161.9	100.1	0.05	0.18	172	30.94	7.6	0.18	No	No
10.2	11	91	2.50	187.0	111.5	0.04	0.18	172	27.30	7.6	0.18	No	No
11.8	7	76	2.50	217.5	126.3	0.04	0.18	172	28.49	7.6	0.18	Yes	Yes
13.2	8	98	2.50	245.4	140.4	0.03	0.17	172	31.01	7.6	0.18	No	No
14.8	5	98	2.50	273.4	152.8	0.03	0.16	172	36.29	7.6	0.18	No	No
16.2	13	74	2.50	300.2	165.8	0.03	0.16	172	32.49	7.6	0.18	No	No
17.8	12	91	2.50	332.6	182.5	0.03	0.15	172	32.39	7.6	0.18	No	No
19.2	25	22	2.50	358.2	194.4	0.05	0.14	241	38.00	7.6	0.18	No	No
2.0	7	93	1.78	30.0	27.9	0.16	0.13	158	34.40	7.6	0.18	No	No
3.2	5	70	1.78	52.9	39.0	0.10	0.16	158	26.00	7.6	0.18	Yes	Yes
4.2	5	99	1.78	72.0	48.2	0.08	0.17	158	26.00	7.6	0.18	Yes	Yes
5.8	6	90	1.78	100.7	61.2	0.06	0.19	158	26.93	7.6	0.18	Yes	Yes
10.2	14	28	1.78	181.2	98.6	0.05	0.20	182	31.49	7.6	0.18	No	No
11.8	14	98	1.78	212.3	114.0	0.04	0.19	167	32.00	7.6	0.18	No	No
13.2	18	15	1.78	238.1	126.1	0.07	0.18	250	33.86	7.6	0.18	No	No
14.8	17	18	1.78	270.7	142.9	0.07	0.17	250	33.86	7.6	0.18	No	No
16.2	17	13	1.78	299.4	157.9	0.06	0.16	250	34.28	7.6	0.18	No	No
17.8	10	75	1.78	331.9	174.7	0.04	0.15	200	34.58	7.6	0.18	No	No
19.2	9	96	1.78	360.4	189.5	0.04	0.14	200	31.97	7.6	0.18	No	No
1.2	6	85	0.85	17.0	13.6	0.27	0.15	161	32.01	7.6	0.18	No	No
3.0	13	99	0.85	49.3	28.2	0.11	0.20	161	29.17	7.6	0.18	No	No
4.8	11	99	0.85	81.6	42.9	0.07	0.22	161	31.49	7.6	0.18	No	No
6.5	4	55	0.85	112.4	57.0	0.05	0.22	161	26.00	7.6	0.18	Yes	Yes
7.5	7	56	0.85	131.3	66.0	0.05	0.22	161	27.75	7.6	0.18	Yes	Yes
9.2	16	50	0.85	163.2	81.3	0.04	0.22	161	32.00	7.6	0.18	No	No
12.5	11	98	0.85	228.2	113.9	0.03	0.20	161	32.01	7.6	0.18	No	No
13.2	7	99	0.85	241.1	119.9	0.03	0.20	161	32.09	7.6	0.18	No	No
14.8	10	88	0.85	269.5	132.6	0.04	0.19	200	32.85	7.6	0.18	No	No
16.2	10	89	0.85	297.2	146.6	0.06	0.17	243	32.62	7.6	0.18	No	No
17.8	16	12	0.85	328.7	162.4	0.06	0.16	243	33.86	7.6	0.18	No	No
1.2	4	7	0.71	20.7	15.9	0.21	0.15	140	24.90	7.6	0.18	Yes	Yes
2.8	6	99	0.71	52.1	31.6	0.09	0.19	140	31.80	7.6	0.18	No	No
4.2	15	51	0.71	77.7	43.4	0.06	0.21	140	30.39	7.6	0.18	No	No
7.0	13	99	0.71	130.4	68.7	0.04	0.21	140	28.59	7.6	0.18	No	No
8.8	8	84	0.71	163.9	84.6	0.03	0.21	140	36.80	7.6	0.18	No	No
11.8	7	68	0.71	223.4	114.6	0.05	0.20	200	28.49	7.6	0.18	Yes	Yes
13.2	10	59	0.71	248.9	126.4	0.04	0.19	184	30.39	7.6	0.18	Yes	Yes
14.8	12	11	0.71	279.5	141.2	0.04	0.18	184	31.49	7.6	0.18	Yes	Yes
16.5	8	91	0.71	313.4	158.5	0.03	0.17	184	34.34	7.6	0.18	No	No
17.8	13	98	0.71	339.6	172.0	0.03	0.16	184	30.00	7.6	0.18	No	No
19.2	8	100	0.71	365.9	184.5	0.03	0.15	184	30.34	7.6	0.18	No	No
1.2	6	8	1.70	17.0	17.0	0.19	0.12	133	26.00	7.6	0.18	No	No
2.8	8	95	1.70	45.4	34.6	0.08	0.15	133	32.85	7.6	0.18	No	No
5.0	8	65	1.70	86.1	53.7	0.05	0.18	133	33.04	7.6	0.18	No	No
6.1	14	14	1.70	106.9	63.7	0.06	0.19	167	30.95	7.6	0.18	No	No
7.2	15	23	1.70	130.9	77.0	0.05	0.19	167	31.49	7.6	0.18	No	No
9.0	6	98	1.70	168.0	96.4	0.06	0.19	200	31.08	7.6	0.18	No	No
10.2	8	97	1.70	189.3	105.9	0.06	0.19	200	32.85	7.6	0.18	No	No
11.8	10	29	1.70	218.8	119.7	0.05	0.19	200	30.39	7.6	0.18	Yes	Yes
14.0	18	6	1.70	261.3	140.6	0.06	0.18	226	34.28	7.6	0.18	No	No
14.8	20	14	1.70	277.8	149.3	0.05	0.17	226	35.10	7.6	0.18	No	No
17.2	18	6	1.70	328.2	176.2	0.05	0.15	226	34.70	7.6	0.18	No	No
19.2	15	90	1.70	373.2	201.6	0.05	0.14	231	31.53	7.6	0.18	No	No
1.2	13	43	2.30	20.5	20.5	0.31	0.12	171	29.17	7.6	0.18	No	No
2.8	7	73	2.30	50.1	45.2	0.13	0.13	171	32.57	7.6	0.18	No	No
4.2	9	54	2.30	76.9	58.3	0.09	0.15	171	28.49	7.6	0.18	No	No
5.8	10	94	2.30	109.3	75.0	0.07	0.17	171	29.17	7.6	0.18	No	No
7.2	9	98	2.30	134.5	86.5	0.06	0.17	171	32.66	7.6	0.18	No	No
8.8	3	99	2.30	164.1	100.3	0.05	0.18	182	32.90	7.6	0.18	No	No
10.2	8	98	2.30	190.2	112.7	0.05	0.18	182	33.09	7.6	0.18	No	No

Table A1 (*continued*)

Z (m)	$(N_1)_{60}$ (—)	$F \leqslant 75\,\mu m$ (%)	d_w (m)	σ_{vo} (kPa)	σ'_{vo} (kPa)	a_t (g)	τ_{av}/σ'_{vo} (—)	V_s (m/s)	ϕ' (°)	M_v (—)	a_{max} (g)	Liquefaction (—)	GRNN (—)
11.8	7	64	2.30	219.1	125.9	0.04	0.18	182	28.49	7.6	0.18	Yes	Yes
13.2	11	17	2.30	247.5	140.6	0.06	0.17	226	30.95	7.6	0.18	Yes	Yes
14.8	14	17	2.30	279.2	156.6	0.05	0.16	226	32.96	7.6	0.18	No	No
16.2	30	20	2.30	308.4	172.0	0.05	0.15	226	39.29	7.6	0.18	No	No
17.8	16	14	2.30	345.3	193.3	0.05	0.14	226	34.28	7.6	0.18	No	No
19.2	13	10	2.30	375.0	209.2	0.05	0.14	231	33.42	7.6	0.18	No	No
2.8	12	96	0.80	45.6	25.9	0.10	0.20	145	33.45	7.6	0.18	No	No
4.2	7	96	0.80	70.3	37.0	0.07	0.22	145	32.09	7.6	0.18	No	No
5.8	9	57	0.80	98.2	49.1	0.05	0.23	145	28.49	7.6	0.18	Yes	Yes
7.2	6	98	0.80	125.5	62.7	0.04	0.23	145	33.14	7.6	0.18	No	No
8.8	13	99	0.80	156.6	78.2	0.03	0.22	145	29.97	7.6	0.18	No	No
10.2	14	97	0.80	182.1	89.9	0.05	0.22	194	31.49	7.6	0.18	No	No
11.8	15	13	0.80	211.1	103.2	0.05	0.21	194	32.00	7.6	0.18	Yes	Yes
14.8	12	91	0.80	271.2	133.8	0.03	0.18	164	35.45	7.6	0.18	No	No
17.8	14	91	0.80	331.1	164.4	0.03	0.16	176	36.80	7.6	0.18	No	No
19.2	13	60	0.80	358.7	178.2	0.06	0.15	250	37.54	7.6	0.18	No	No
2.2	9	15	2.00	39.1	37.1	0.13	0.12	150	28.49	7.6	0.18	Yes	Yes
4.2	4	99	2.00	80.9	59.3	0.13	0.16	200	26.00	7.6	0.18	Yes	Yes
5.8	6	54	2.00	110.2	73.0	0.06	0.17	158	26.93	7.6	0.18	Yes	Yes
7.2	14	9	2.00	137.5	86.4	0.05	0.18	158	31.49	7.6	0.18	Yes	Yes[a]
8.8	7	96	2.00	171.0	104.3	0.04	0.18	154	32.01	7.6	0.18	No	No
10.2	15	34	2.00	197.1	116.6	0.06	0.18	200	32.49	7.6	0.18	No	No
11.8	23	18	2.00	228.5	132.3	0.05	0.18	200	35.87	7.6	0.18	No	No
13.2	25	7	2.00	257.4	147.6	0.05	0.17	200	36.97	7.6	0.18	No	No
14.8	26	20	2.00	292.1	166.5	0.04	0.16	200	37.66	7.6	0.18	No	No
16.2	17	10	2.00	323.4	184.1	0.04	0.15	200	34.70	7.6	0.18	No	No
17.8	21	6	2.00	358.3	203.3	0.04	0.14	200	36.61	7.6	0.18	No	No
19.2	10	29	2.00	388.9	220.2	0.06	0.13	250	31.49	7.6	0.18	No	No
2.2	7	59	1.10	36.4	25.7	0.08	0.17	118	26.93	7.6	0.18	Yes	Yes
3.8	10	17	1.10	66.5	40.0	0.05	0.19	118	28.49	7.6	0.18	Yes	Yes
5.2	9	61	1.10	93.6	53.4	0.06	0.20	150	28.49	7.6	0.18	Yes	Yes
6.8	10	15	1.10	124.0	68.1	0.05	0.21	150	29.17	7.6	0.18	Yes	Yes
8.2	16	15	1.10	153.5	83.9	0.05	0.20	167	32.00	7.6	0.18	Yes	Yes
9.8	13	98	1.10	186.9	101.6	0.04	0.20	167	31.34	7.6	0.18	No	No
11.2	20	20	1.10	212.2	113.1	0.07	0.20	235	34.28	7.6	0.18	No	No
12.8	25	7	1.10	244.1	129.3	0.07	0.19	235	36.25	7.6	0.18	No	No
14.2	19	18	1.10	275.1	146.6	0.06	0.18	235	34.70	7.6	0.18	No	No
15.8	24	18	1.10	308.0	163.8	0.06	0.16	235	36.97	7.6	0.18	No	No
17.2	21	10	1.10	339.4	181.4	0.05	0.15	235	36.25	7.6	0.18	No	No
18.8	16	98	1.10	374.6	201.0	0.06	0.14	250	30.09	7.6	0.18	No	No
1.2	7	85	2.30	17.7	17.7	0.18	0.12	130	29.01	7.6	0.18	No	No
2.8	6	30	2.30	43.5	38.6	0.08	0.13	130	26.93	7.6	0.18	Yes	Yes
4.2	5	90	2.30	70.4	51.8	0.07	0.16	147	26.00	7.6	0.18	Yes	Yes
6.8	6	99	2.30	118.2	74.1	0.04	0.18	147	27.35	7.6	0.18	Yes	Yes
8.8	7	94	2.30	156.3	92.6	0.03	0.19	147	28.12	7.6	0.18	Yes	Yes
10.1	18	23	2.30	180.5	103.9	0.01	0.19	97	33.19	7.6	0.18	No	No
11.8	13	31	2.30	213.8	120.6	0.01	0.18	97	31.49	7.6	0.18	No	No
13.2	11	94	2.30	241.0	134.1	0.02	0.18	133	34.77	7.6	0.18	No	No
14.8	18	67	2.30	272.3	149.7	0.02	0.17	133	34.28	7.6	0.18	No	No
16.2	11	30	2.30	300.7	164.3	0.02	0.16	133	31.22	7.6	0.18	No	No[a]
17.8	18	20	2.30	331.9	179.9	0.02	0.15	133	35.10	7.6	0.18	No	No
19.2	43	29	2.30	361.1	195.3	0.02	0.14	133	43.75	7.6	0.18	No	No
1.2	25	81	2.30	16.4	16.4	0.15	0.12	120	34.34	7.6	0.18	No	No
2.8	5	99	2.30	40.1	35.1	0.07	0.13	120	27.34	7.6	0.18	No	No
4.2	6	19	2.30	64.4	45.7	0.09	0.16	167	26.93	7.6	0.18	Yes	Yes
5.8	7	99	2.30	96.1	61.8	0.06	0.18	159	30.19	7.6	0.18	No	No
7.2	7	99	2.30	121.2	73.1	0.05	0.19	159	31.72	7.6	0.18	No	No
8.8	7	99	2.30	150.1	86.4	0.04	0.19	159	30.74	7.6	0.18	No	No
10.2	6	99	2.30	175.3	97.8	0.04	0.19	159	30.41	7.6	0.18	No	No
11.8	8	66	2.30	203.1	110.0	0.03	0.19	159	29.17	7.6	0.18	Yes	Yes
13.2	21	10	2.30	229.7	122.8	0.05	0.18	207	34.70	7.6	0.18	No	No
14.8	19	8	2.30	262.4	139.8	0.05	0.17	207	34.70	7.6	0.18	No	No
16.2	16	7	2.30	291.6	155.3	0.04	0.16	207	33.86	7.6	0.18	No	No

Table A1 (*continued*)

Z (m)	$(N_1)_{60}$ (—)	$F \leqslant 75\,\mu m$ (%)	d_w (m)	σ_{vo} (kPa)	σ'_{vo} (kPa)	a_t (g)	τ_{av}/σ'_{vo} (—)	V_s (m/s)	ϕ' (°)	M_v (—)	a_{max} (g)	Liquefaction (—)	GRNN (—)
17.2	8	82	2.30	313.0	166.8	0.04	0.15	207	30.10	7.6	0.18	No	No
19.2	10	97	2.30	351.7	185.9	0.05	0.14	235	31.92	7.6	0.18	No	No
2.8	13	80	0.99	50.9	33.2	0.06	0.38	113	29.80	7.6	0.38	Yes	Yes
4.2	8	80	0.99	78.6	47.1	0.08	0.41	155	27.75	7.6	0.38	Yes	Yes
5.4	13	90	0.99	101.8	58.5	0.09	0.42	194	30.39	7.6	0.38	Yes	Yes
7.2	15	90	0.99	138.0	77.1	0.08	0.43	205	31.49	7.6	0.38	Yes	Yes
9.2	15	82	0.99	179.3	98.7	0.06	0.42	188	32.00	7.6	0.38	Yes	Yes
11.2	16	90	0.99	219.5	119.3	0.06	0.40	206	32.96	7.6	0.38	Yes	Yes
12.2	18	95	0.99	239.3	129.3	0.05	0.39	206	36.88	7.6	0.38	No	No
13.8	17	88	0.99	271.8	146.1	0.07	0.37	242	33.86	7.6	0.38	Yes	Yes
15.2	21	78	0.99	299.5	160.1	0.08	0.35	281	35.87	7.6	0.38	No	No
17.2	30	20	0.99	339.1	180.0	0.10	0.33	315	39.29	7.6	0.38	No	No
1.2	30	43	0.70	39.7	34.8	0.23	0.28	149	35.10	7.6	0.38	No	No
3.2	15	10	0.70	84.1	59.6	0.05	0.35	113	30.95	7.6	0.38	Yes	Yes
6.2	20	32	0.70	150.4	96.4	0.09	0.37	205	33.86	7.6	0.38	Yes	Yes
8.6	13	57	0.70	206.1	128.6	0.06	0.37	195	32.00	7.6	0.38	Yes	Yes
10.2	18	38	0.70	237.1	143.9	0.05	0.37	188	34.28	7.6	0.38	Yes	Yes
12.2	21	19	0.70	279.2	166.3	0.05	0.36	206	35.87	7.6	0.38	Yes	Yes
13.2	33	58	0.70	299.8	177.2	0.07	0.35	242	40.20	7.6	0.38	No	No
14.2	36	19	0.70	320.6	188.2	0.06	0.34	242	41.35	7.6	0.38	No	No
14.8	62	11	0.70	333.6	195.3	0.08	0.33	281	48.35	7.6	0.38	No	No
16.2	49	26	0.70	362.6	210.5	0.08	0.31	281	45.69	7.6	0.38	No	No
1.2	41	22	5.00	24.8	24.8	0.27	0.25	161	36.97	7.6	0.38	No	No
4.2	10	42	5.00	67.6	67.6	0.08	0.24	161	29.17	7.6	0.38	No	No
5.8	9	25	5.00	96.8	89.0	0.08	0.26	184	29.17	7.6	0.38	Yes	Yes
7.2	23	19	5.00	125.1	103.5	0.07	0.29	184	35.10	7.6	0.38	Yes	Yes
8.8	14	43	5.00	161.1	123.8	0.05	0.30	184	32.00	7.6	0.38	Yes	Yes
10.2	18	55	5.00	189.6	138.6	0.05	0.31	184	36.37	7.6	0.38	No	No
11.8	13	85	5.00	221.9	155.2	0.04	0.31	184	36.71	7.6	0.38	No	No
13.2	17	80	5.00	250.4	169.9	0.07	0.30	250	42.00	7.6	0.38	No	No
14.8	18	97	5.00	282.2	186.1	0.07	0.29	250	52.08	7.6	0.38	No	No
17.8	31	80	5.00	342.5	217.0	0.08	0.27	294	40.49	7.6	0.38	No	No
19.2	38	29	5.00	371.3	232.0	0.08	0.26	294	43.24	7.6	0.38	No	No
1.2	27	13	1.50	21.5	21.5	0.24	0.25	150	33.42	7.6	0.38	No	No
3.8	15	30	1.50	74.1	51.5	0.08	0.35	150	30.95	7.6	0.38	Yes	Yes
5.8	14	30	1.50	115.1	73.0	0.06	0.38	160	30.95	7.6	0.38	Yes	Yes
7.2	22	6	1.50	144.2	88.3	0.05	0.39	160	34.28	7.6	0.38	Yes	Yes
10.2	5	73	1.50	208.0	122.7	0.08	0.38	240	31.34	7.6	0.38	No	No
10.8	4	97	1.50	220.2	129.0	0.08	0.38	240	32.48	7.6	0.38	No	No
11.8	16	98	1.50	240.3	139.3	0.07	0.37	240	31.23	7.6	0.38	No	No
13.2	14	96	1.50	267.4	152.7	0.07	0.36	240	30.84	7.6	0.38	No	No
14.8	18	98	1.50	299.3	168.9	0.06	0.34	240	34.58	7.6	0.38	No	No
16.2	18	99	1.50	326.4	182.2	0.05	0.32	214	32.90	7.6	0.38	No	No
17.8	28	36	1.50	358.2	198.3	0.04	0.30	214	39.29	7.6	0.38	No	No
18.8	63	16	1.50	378.9	209.2	0.04	0.29	214	48.98	7.6	0.38	No	No
7.2	6	94	1.90	147.5	95.5	0.15	0.37	277	34.71	7.6	0.38	No	No
8.8	63	67	1.90	179.4	111.7	0.12	0.37	277	34.90	7.6	0.38	No	No
12.0	14	75	1.90	245.5	146.4	0.17	0.36	363	35.10	7.6	0.38	No	No
13.2	42	42	1.90	269.7	158.9	0.15	0.35	363	42.18	7.6	0.38	No	No
14.8	36	35	1.90	302.7	176.2	0.14	0.33	363	41.07	7.6	0.38	No	No
16.2	30	91	1.90	332.1	191.8	0.13	0.31	363	39.60	7.6	0.38	No	No
17.8	33	49	1.90	366.0	210.1	0.12	0.29	363	41.07	7.6	0.38	No	No
1.8	4	98	1.14	29.7	23.2	0.30	0.56	208	24.90	7.6	0.67	Yes	Yes
3.2	14	51	1.14	56.9	36.7	0.06	0.67	122	33.77	7.6	0.67	No	No
4.2	8	32	1.14	76.7	46.7	0.05	0.71	130	27.75	7.6	0.67	Yes	Yes
5.2	21	46	1.14	96.1	56.3	0.05	0.73	145	32.96	7.6	0.67	Yes	Yes
6.2	16	15	1.14	114.1	64.5	0.08	0.75	188	31.49	7.6	0.67	Yes	Yes
8.6	18	29	1.14	163.2	90.0	0.07	0.74	209	32.96	7.6	0.67	Yes	Yes
11.2	23	33	1.14	216.7	118.0	0.08	0.71	239	35.49	7.6	0.67	No	No
12.2	36	32	1.14	238.2	129.7	0.07	0.69	239	39.60	7.6	0.67	No	No
13.2	28	35	1.14	259.7	141.4	0.08	0.67	261	37.66	7.6	0.67	Yes	Yes
14.2	58	32	1.14	281.5	153.4	0.08	0.64	261	45.92	7.6	0.67	No	No
2.0	10	58	0.75	37.7	25.4	0.27	0.65	208	36.88	7.6	0.67	No	No

Table A1 (*continued*)

Z (m)	$(N_1)_{60}$ (—)	$F \leqslant 75\,\mu m$ (%)	d_w (m)	σ_{vo} (kPa)	σ'_{vo} (kPa)	a_t (g)	τ_{av}/σ'_{vo} (—)	V_s (m/s)	ϕ' (°)	M_v (—)	a_{max} (g)	Liquefaction (—)	GRNN (—)
4.2	9	54	0.75	82.5	48.7	0.05	0.73	130	28.49	7.6	0.67	Yes	Yes
7.2	11	55	0.75	144.6	81.3	0.07	0.74	188	29.80	7.6	0.67	Yes	Yes
10.2	21	52	0.75	206.5	113.8	0.08	0.72	239	34.70	7.6	0.67	Yes	Yes
12.2	24	9	0.75	248.3	135.9	0.07	0.69	239	36.25	7.6	0.67	Yes	Yes
13.2	24	15	0.75	269.7	147.6	0.08	0.66	261	36.61	7.6	0.67	Yes	Yes
14.2	19	34	0.75	291.4	159.4	0.08	0.64	261	35.10	7.6	0.67	Yes	Yes
6.2	13	54	0.50	123.1	67.2	0.08	0.77	188	36.88	7.6	0.67	No	No
8.8	15	51	0.50	176.8	95.4	0.07	0.76	209	32.00	7.6	0.67	Yes	Yes
11.2	13	40	0.50	226.6	121.7	0.08	0.72	239	31.49	7.6	0.67	Yes	Yes
13.2	20	16	0.50	270.2	145.6	0.08	0.67	261	35.10	7.6	0.67	Yes	Yes
14.2	20	21	0.50	291.9	157.5	0.08	0.65	261	35.49	7.6	0.67	Yes	Yes
15.2	15	19	0.50	313.2	169.0	0.07	0.62	260	33.42	7.6	0.67	Yes	Yes
16.2	30	10	0.50	334.5	180.5	0.07	0.59	260	39.29	7.6	0.67	No	No
17.2	28	13	0.50	356.3	192.4	0.07	0.56	262	38.97	7.6	0.67	No	No
18.2	18	5	0.50	378.0	204.4	0.06	0.54	262	35.49	7.6	0.67	Yes	Yes
19.2	32	10	0.50	400.1	216.6	0.07	0.52	268	40.78	7.6	0.67	No	No
2.8	9	31	1.57	55.2	43.1	0.07	0.55	122	28.49	7.6	0.67	Yes	Yes
3.8	18	19	1.57	75.8	54.0	0.06	0.60	130	32.00	7.6	0.67	Yes	Yes
5.6	26	36	1.57	112.9	73.4	0.05	0.65	145	35.10	7.6	0.67	No	No
8.2	19	15	1.57	169.1	104.1	0.08	0.67	209	33.86	7.6	0.67	Yes	Yes
0.8	8	13	3.60	12.1	12.1	0.85	0.44	233	26.93	7.6	0.67	No	No
1.8	8	10	3.60	28.9	28.9	0.30	0.44	208	27.75	7.6	0.67	No	No
2.8	1	13	3.60	43.1	43.1	0.07	0.43	122	23.46	7.6	0.67	No	No
3.8	4	31	3.60	58.9	56.9	0.06	0.45	130	26.00	7.6	0.67	Yes	Yes
5.2	21	10	3.60	87.6	71.9	0.05	0.52	145	33.42	7.6	0.67	Yes	Yes
6.2	14	36	3.60	109.2	83.7	0.08	0.55	188	31.49	7.6	0.67	Yes	Yes
7.2	18	26	3.60	131.4	96.0	0.07	0.57	188	32.96	7.6	0.67	Yes	Yes
8.2	22	33	3.60	153.0	107.9	0.08	0.59	209	34.70	7.6	0.67	Yes	Yes
9.2	41	27	3.60	174.9	120.0	0.09	0.59	239	40.78	7.6	0.67	No	No
10.2	22	26	3.60	197.3	132.5	0.08	0.59	239	35.49	7.6	0.67	Yes	Yes
11.2	24	8	3.60	219.8	145.3	0.08	0.59	239	36.61	7.6	0.67	Yes	Yes
12.2	21	9	3.60	242.2	157.8	0.07	0.58	239	35.87	7.6	0.67	Yes	Yes
13.2	25	67	3.60	263.3	169.2	0.08	0.56	261	37.32	7.6	0.67	No	No
14.2	22	52	3.60	284.6	180.6	0.08	0.55	261	36.61	7.6	0.67	No	No
15.2	15	53	3.60	305.8	192.0	0.07	0.53	260	33.86	7.6	0.67	Yes	Yes
0.8	14	23	1.03	13.1	13.1	0.33	0.44	145	29.17	7.6	0.67	No	No
3.8	16	19	1.03	73.7	46.5	0.11	0.68	177	30.95	7.6	0.67	Yes	Yes
6.8	23	14	1.03	136.7	80.1	0.05	0.72	151	34.28	7.6	0.67	Yes	Yes
7.8	11	10	1.03	157.9	91.5	0.04	0.72	143	30.39	7.6	0.67	Yes	Yes
9.8	15	23	1.03	198.9	112.8	0.03	0.71	149	32.49	7.6	0.67	Yes	Yes
10.8	17	18	1.03	219.8	123.9	0.03	0.70	149	33.42	7.6	0.67	Yes	Yes
15.8	25	18	1.03	323.0	178.1	0.05	0.59	224	37.66	7.6	0.67	Yes	Yes
17.8	12	17	1.03	366.9	202.4	0.06	0.54	245	32.49	7.6	0.67	Yes	Yes
18.8	26	26	1.03	387.9	213.5	0.05	0.52	245	38.65	7.6	0.67	No	No
19.8	28	17	1.03	408.9	224.7	0.06	0.50	265	39.60	7.6	0.67	No	No
1.2	11	49	3.20	19.3	19.3	0.18	0.44	130	28.49	7.6	0.67	No	No
3.8	18	20	3.20	64.6	58.7	0.10	0.47	167	32.00	7.6	0.67	Yes	Yes
5.8	16	50	3.20	108.7	83.2	0.07	0.55	179	32.00	7.6	0.67	Yes	Yes
7.2	16	40	3.20	138.4	99.2	0.06	0.58	179	32.49	7.6	0.67	Yes	Yes
8.8	15	72	3.20	171.8	116.9	0.05	0.60	179	42.81	7.6	0.67	No	No
10.2	13	65	3.20	200.9	132.3	0.05	0.61	179	32.00	7.6	0.67	Yes	Yes
11.8	27	27	3.20	234.0	149.6	0.04	0.60	179	37.66	7.6	0.67	No	No
13.2	30	20	3.20	263.2	165.1	0.04	0.58	179	38.97	7.6	0.67	No	No
14.8	35	49	3.20	296.9	183.1	0.03	0.55	179	41.07	7.6	0.67	No	No
18.8	24	45	3.20	386.7	233.7	0.03	0.47	179	38.65	7.6	0.67	No	No
1.2	3	65	0.50	21.7	14.8	0.11	0.64	104	24.24	7.6	0.67	Yes	Yes
3.8	8	61	0.50	74.0	41.6	0.04	0.77	104	27.75	7.6	0.67	Yes	Yes
5.8	9	75	0.50	115.4	63.4	0.07	0.77	167	36.05	7.6	0.67	No	No
7.5	25	45	0.50	151.7	83.0	0.05	0.76	167	35.10	7.6	0.67	Yes	Yes[a]
8.8	21	44	0.50	179.1	97.7	0.05	0.75	167	34.28	7.6	0.67	Yes	Yes
10.2	24	51	0.50	208.5	113.4	0.04	0.73	167	35.49	7.6	0.67	No	No
11.8	27	58	0.50	242.3	131.4	0.04	0.70	167	36.97	7.6	0.67	No	No
13.2	17	77	0.50	270.8	146.3	0.03	0.67	167	36.97	7.6	0.67	No	No

Table A1 (*continued*)

Z (m)	$(N_1)_{60}$ (—)	$F \leqslant 75\,\mu m$ (%)	d_w (m)	σ_{vo} (kPa)	σ'_{vo} (kPa)	a_t (g)	τ_{av}/σ'_{vo} (—)	V_s (m/s)	ϕ' (°)	M_v (—)	a_{max} (g)	Liquefaction (—)	GRNN (—)
14.8	38	29	0.50	304.7	164.4	0.03	0.63	167	41.35	7.6	0.67	No	No
16.2	13	72	0.50	334.1	180.1	0.06	0.59	250	36.97	7.6	0.67	No	No
17.8	13	76	0.50	367.5	197.7	0.06	0.55	250	34.97	7.6	0.67	No	No
19.2	11	82	0.50	396.8	213.3	0.10	0.52	333	34.46	7.6	0.67	No	No
1.2	27	36	1.40	20.4	20.4	0.18	0.44	130	33.42	7.6	0.67	No	No
3.8	14	69	1.40	70.3	46.7	0.06	0.65	130	30.39	7.6	0.67	Yes	Yes
5.8	16	22	1.40	109.3	66.2	0.04	0.70	130	31.49	7.6	0.67	Yes	Yes
7.2	19	70	1.40	139.1	82.2	0.08	0.71	200	32.96	7.6	0.67	Yes	Yes
8.8	28	71	1.40	173.1	100.5	0.05	0.71	182	37.25	7.6	0.67	No	No
10.2	21	71	1.40	202.9	116.6	0.05	0.69	182	34.70	7.6	0.67	Yes	Yes
11.8	26	51	1.40	236.6	134.6	0.04	0.67	182	36.97	7.6	0.67	No	No
13.2	18	74	1.40	266.4	150.6	0.04	0.64	182	34.10	7.6	0.67	No	No
14.8	23	40	1.40	299.3	167.8	0.04	0.61	182	36.61	7.6	0.67	No	No
16.2	37	35	1.40	328.6	183.5	0.03	0.57	182	41.63	7.6	0.67	No	No
17.8	16	67	1.40	362.9	202.0	0.03	0.53	182	38.27	7.6	0.67	No	No
19.2	17	67	1.40	391.8	217.2	0.03	0.51	182	37.54	7.6	0.67	No	No

[a]These case records have errors either by wrong prediction or exceedance of error limit.

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